
Software Requirements and Design Document

for

PawCare

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VentiTech

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PawCare - Animal Rescue and Care System

1. Introduction:

1.1 Purpose

This Software Requirements Specification (SRS) document defines the requirements for the PawCare -Animal Rescue and Care System. The system is designed to assist users in addressing the challenges of helping injured or lost animals by providing tools for managing reports, offering first aid guidance, and connecting with rescue centers

1.2 Scope

This project targets individuals who encounter injured or lost animals, providing immediate access to nearby animal care centers, vets, and first aid information.

The application will serve communities in densely populated areas where animal accidents or neglect are more common. While no exact tool currently exists, an article mentioned the planned launch of a similar app in Pakistan in 2020(Source), but it still has not been released, highlighting a gap that this project aims to fill.

1.3 Title

PawCare - Animal Rescue and Care System.

1.4 Objective

- To provide a platform that locates nearby animal care centres, vets, and offers immediate first aid guidance based on the situation described.
- To connect treated animals with rescue centers that can facilitate their adoption or placement in foster care.
- To include a missing pet page to help identify if reported animals are lost pets and reconnect them with their owners.
- To enable volunteers to register for transporting animals to care facilities, enhancing community involvement in animal rescue efforts.
- To integrate a quick assistant for providing fast and relevant first aid information.

1.5 Problem Statement and Description

Animals often suffer due to accidents or abandonment in populated regions, with limited immediate help available. Many individuals who wish to assist are hindered by a lack of resources or knowledge, resulting in extended suffering for the animals. Additionally, there is often no streamlined process to reconnect lost pets with their owners or find homes for rescued animals.

This project seeks to address these challenges by offering a comprehensive solution that provides first aid guidance, locates nearby care facilities, and ensures animals receive continued care through rescue centers, where they can be adopted or fostered.

The inclusion of a missing pet page aids in reuniting pets with their families, and the volunteer program encourages community engagement in the rescue process.

2. Overall Description

2.1 Product Perspective

PawCare is a new, self-contained product designed specifically to address the lack of a streamlined solution for animal rescue and care in urban areas. Unlike any existing tools, it fills a gap in the market where no similar system currently provides combined functionalities for locating care centers, offering first aid guidance, and reconnecting lost pets with owners.

2.2 Product Functions

- Users can report missing animals
- Locates nearby rescue centers and users can send them alerts
- Provides first aid guidance to users
- Users can find their missing animals
- Users can adopt recovered animals
- Users can make donations to rescue centers
- Users can register as volunteer for the system
- Rescue center handles all the requests from users
- Rescue centers can contact registered vets
- Vets can generate injury reports for rescue centers

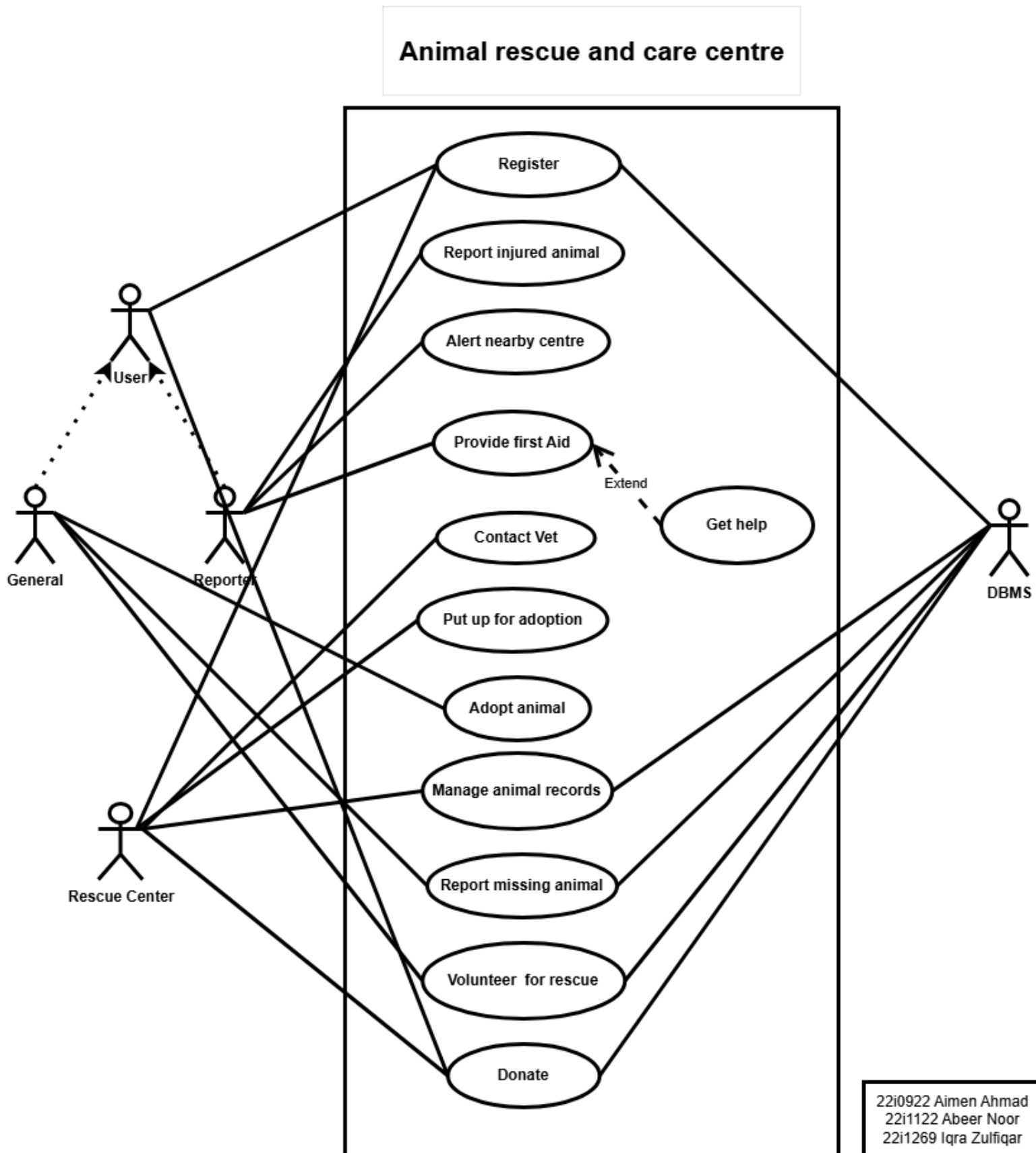
2.3 List of Use Cases

- Register
- Report Injured Animal
- Alert Nearby Centers
- Provide First Aid
- Contact Vet
- Put up for adoption
- Adopt Animal
- Manage Animal Records
- Report Missing Animal
- Volunteer for rescue
- Donate

2.4 Extended Use Cases

Get Help

2.5 Use Case Diagram



3. Other Nonfunctional Requirements

3.1 Performance Requirements

The system must meet the following performance requirements to ensure its efficiency and usability under various circumstances:

- Geolocation Services: The system should locate nearby animal care centers.
- First Aid Guidance: The first aid module must provide relevant instructions based on user input quickly without delay.
- Lost Pet Search: Searching for a lost pet in the system's database should return results promptly.

These requirements are critical for ensuring user satisfaction and maintaining the platform's reliability in real-time scenarios.

3.2 Safety Requirements

These following measures will ensure the system prioritizes the safety of both users and animals while maintaining ethical and legal compliance.

Accurate First Aid Guidance:

- The system must ensure that all first aid suggestions are verified by veterinary experts to prevent harm to animals due to incorrect or incomplete instructions.
- Users must acknowledge a disclaimer indicating that first aid is a temporary solution and professional care should be sought immediately.

User Safety:

- Volunteers assisting in rescue operations must be advised on handling injured animals safely to avoid personal harm or further injury to the animal.

Compliance with Legal Standards:

- The system must adhere to local animal welfare regulations, ensuring no actions facilitated by the platform contradict animal protection laws.
- Any data sharing with third parties (e.g., rescue centers, veterinarians) must comply with privacy and data protection regulations.

3.3 Security Requirements

The following requirements ensure that the system protects user data, maintains confidentiality, and resists unauthorized access.

User Authentication and Authorization:

- All users, including volunteers, rescue centers, and pet owners, must authenticate their identities using secure login method.
- Access to sensitive functionalities, such as volunteer coordination and lost pet data, must be restricted to authorized users.

3.4 Software Quality Attributes

Usability:

- The interface must be user-friendly, enabling non-technical users to perform key actions such as locating care centers or submitting reports with minimal effort.
- All functions must be accessible with no more than three clicks from the home screen.

Reliability:

- The system must operate without failure for 99.9% of the time during a calendar month.
- Critical functionalities, such as geolocation and first aid guidance, must be available 24/7.

Scalability:

- The system must handle an increase in user base without performance degradation

Interoperability:

- The system must integrate seamlessly with third-party APIs like Open Street Maps and external databases.
- Standardized data formats (e.g., JSON or XML) should be used to facilitate communication between components.

Correctness:

- All outputs, including first aid instructions and geolocation results, must have an accuracy rate of at least 95%.

Robustness:

- The system must handle invalid inputs gracefully, providing clear error messages and suggestions for corrective actions.

3.5 Business Rules

These rules ensure the system operates efficiently, securely, and ethically, aligning with its objectives to promote animal welfare and community engagement.

Role-Specific Access:

- Volunteers can register and view tasks assigned to them but cannot access the personal details of pet owners or rescue center staff.

Adoption Eligibility:

- Individuals interested in adopting animals must complete a mandatory application form that is reviewed by the rescue center before approval.

Volunteer Verification:

- All volunteers must be verified before they can assist in transportation or rescue operations.

3.6 Operating Environment

The below environment specifications ensures the desktop application performs effectively while meeting user and system requirements.

Hardware Platform:

- The application must run on desktops or laptops with a minimum configuration of 4GB RAM, 500GB storage, and a dual-core processor.

Operating System:

- The software is compatible with Windows 10 (or later) and macOS Catalina (or later).

Software Dependencies:

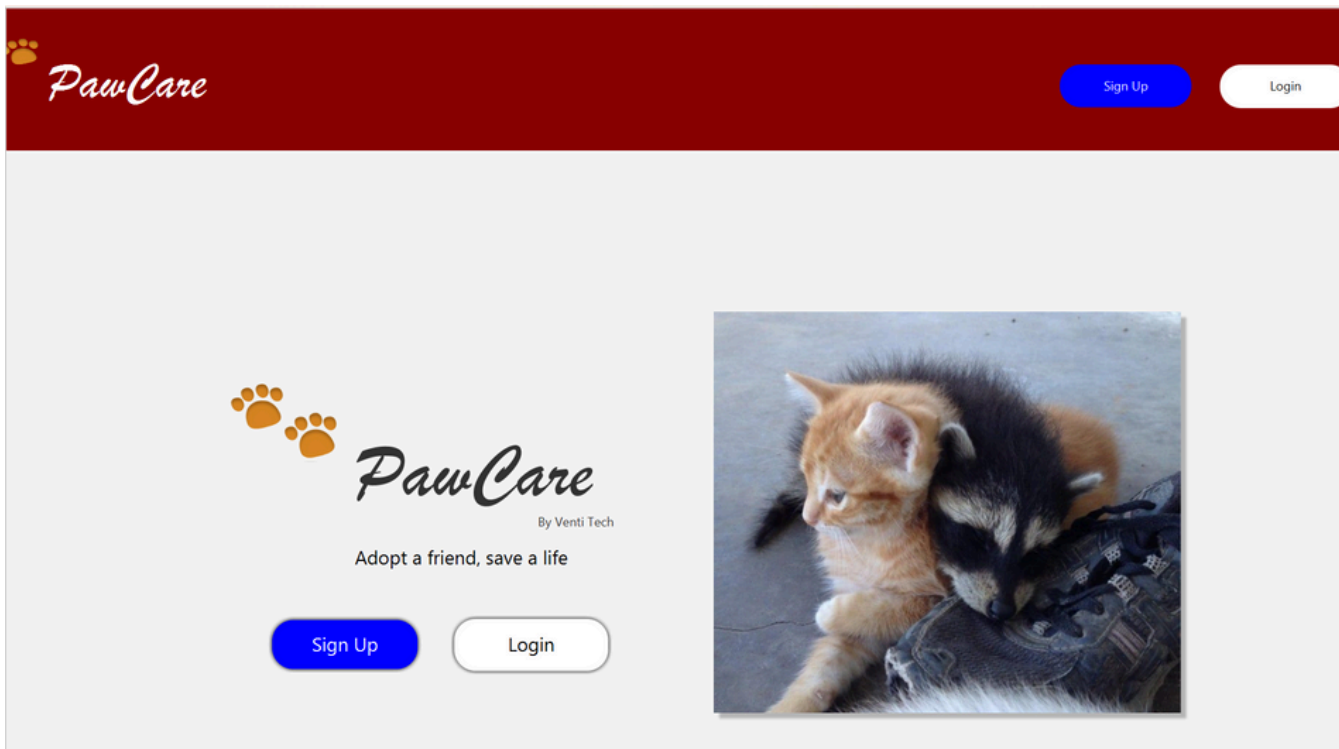
- The application requires Java Runtime Environment (JRE) version 11 or higher for execution.
- It relies on JDBC for database connectivity to manage data storage and retrieval.

Database Environment:

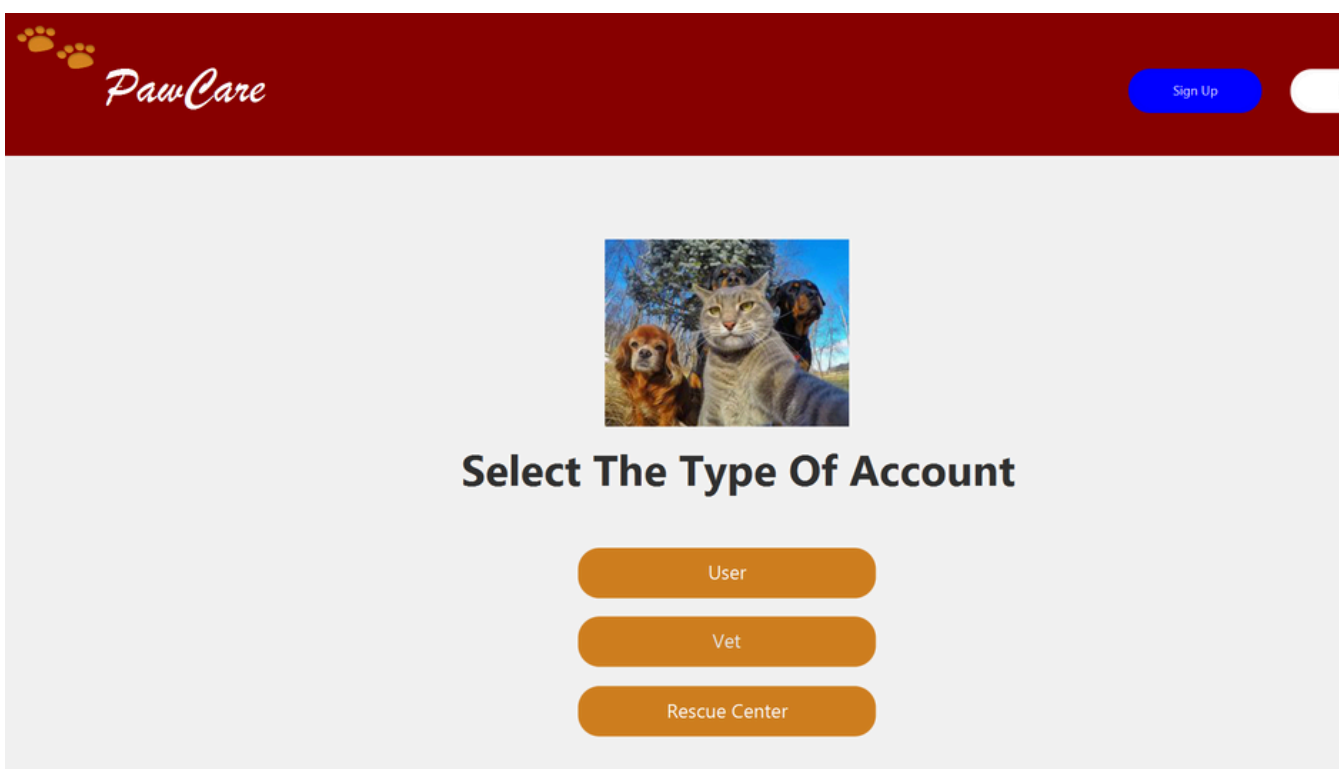
- The application connects to a local or remote database server for managing user data, reports, and rescue center information.

7 User Interfaces

Following are the interfaces of our app. The Program opens up to the page:



1. Sign up:





Sign Up

Logi

User Sign up

Fill in the following information to sign up

☒ I have read and confirm the Terms of Agreement

Sign Up





Sign Up

Lo

Rescue Center Sign up

Fill in the following information to sign up

☒ I have read and confirm the Terms of Agreement

Sign Up





Sign Up

Log

Vet Sign up

Fill in the following information to sign up

Username

Name

Email

Password

Vet Center Location

Phone Number

☒ I have read and confirm the Terms of Agreement

Sign Up



2. Login



Sign Up

Login

Login

Fill in the following information to sign up

Enter Password

☐ I have read and confirm the [Terms of Agreement](#)

User

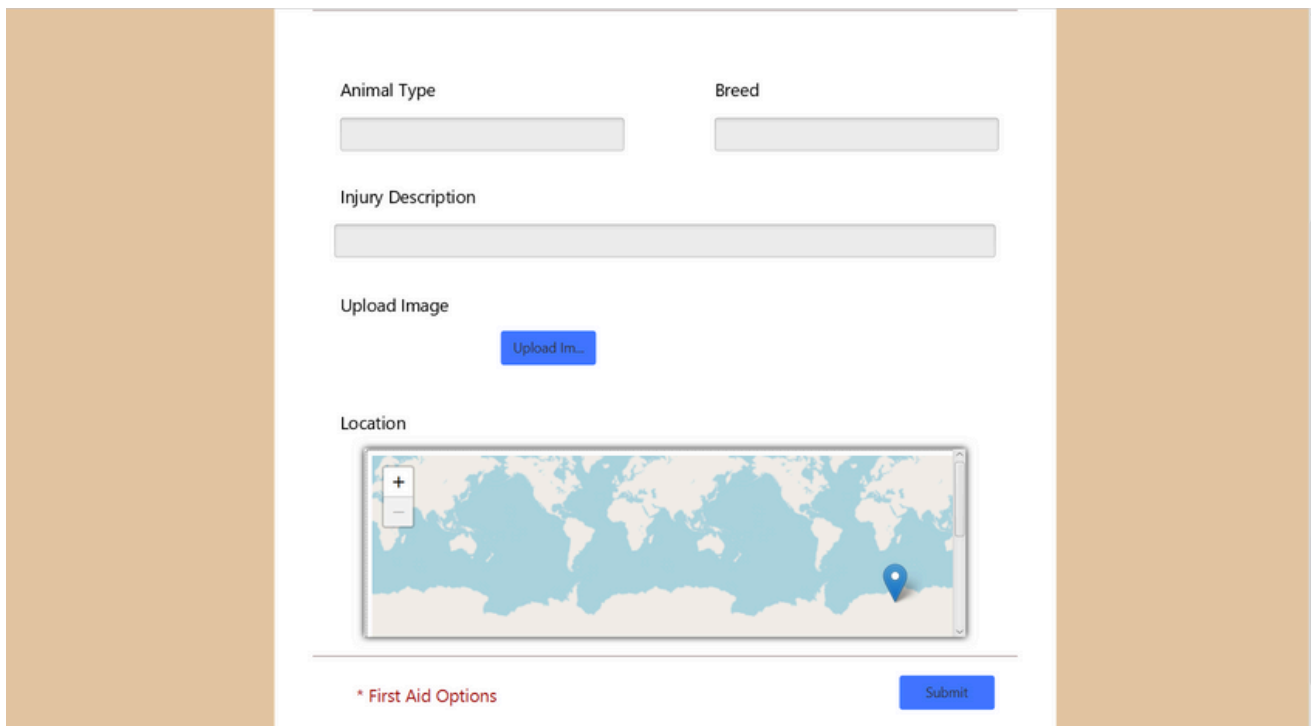
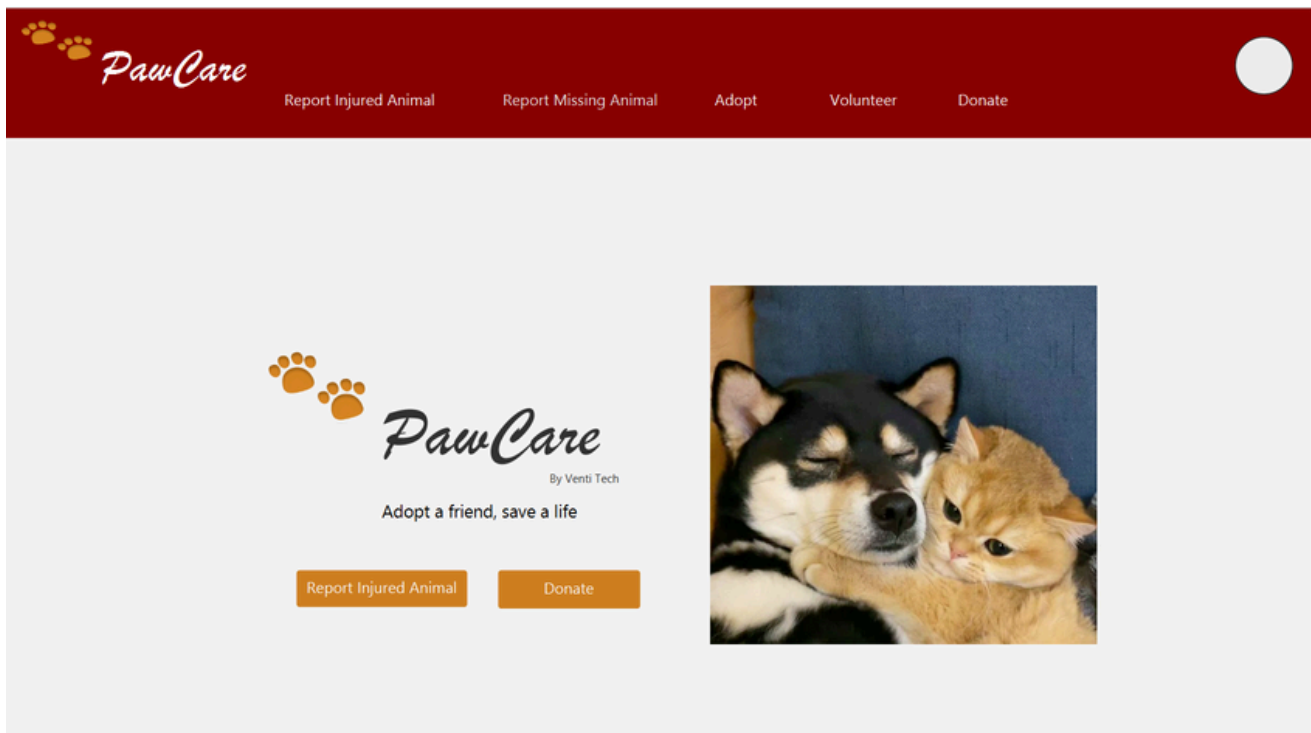
Vet

Rescue Center

Login

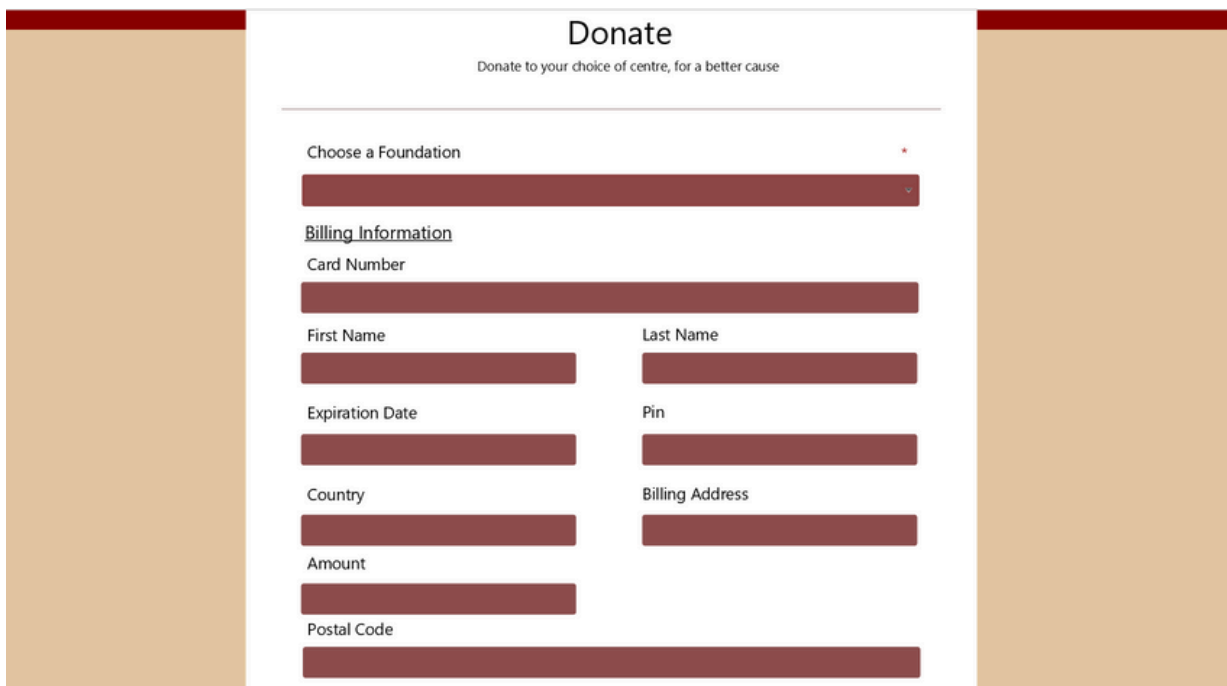
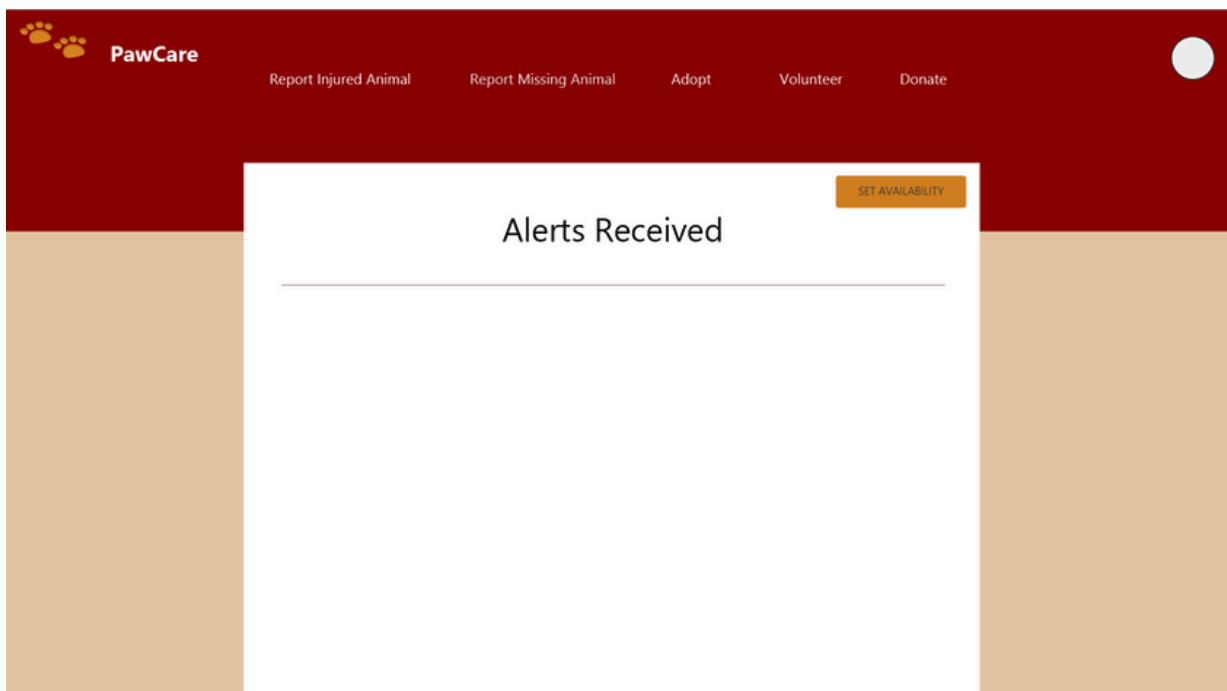
3. User interface:

The following are the different pages for the user

The image shows a form for reporting an animal. The form is centered on a white background with tan sidebars. It contains the following fields and elements: "Animal Type" and "Breed" are text input fields. "Injury Description" is a larger text input field. "Upload Image" is a label above a blue "Upload Image" button. "Location" is a label above a map input field. At the bottom left, there is a red text label "* First Aid Options". At the bottom right, there is a blue "Submit" button.

Report Missing Animal

Volunteer



Donate

Enter Phone Number

Billing Information

Select Rescue Center

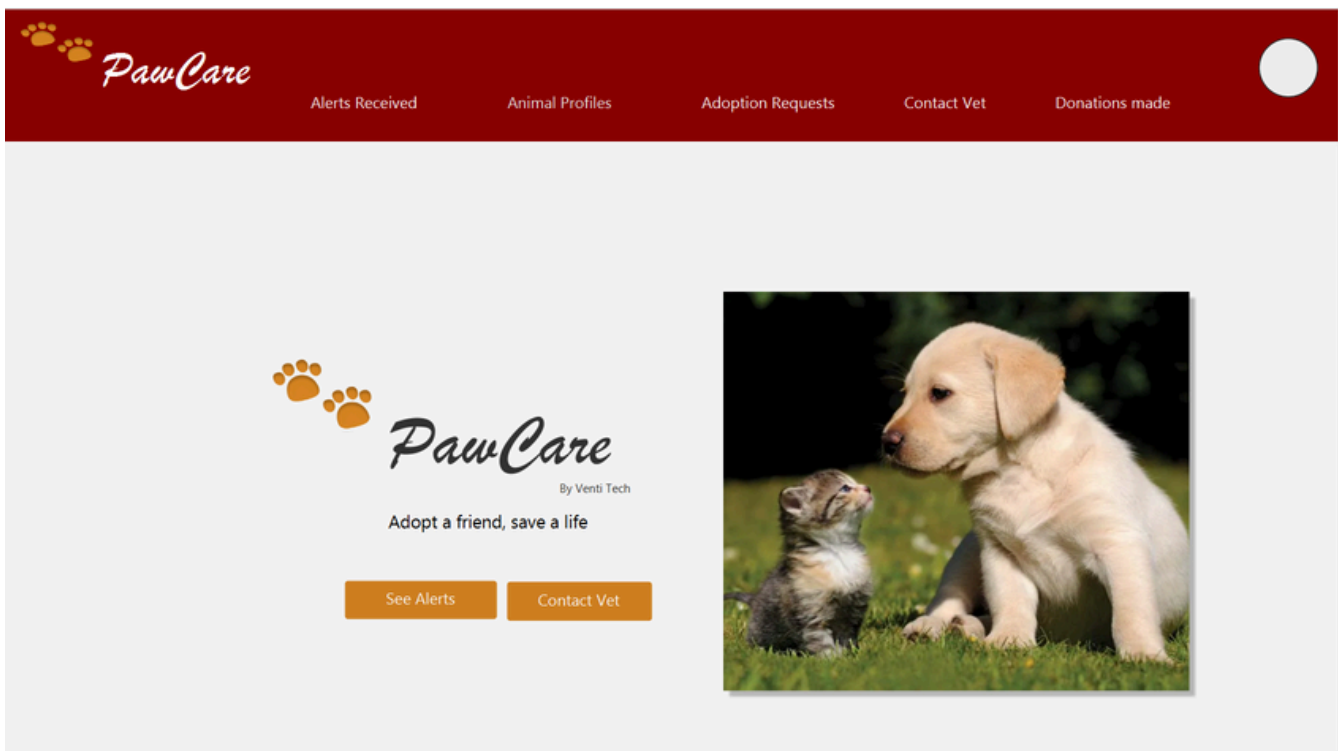
First Name

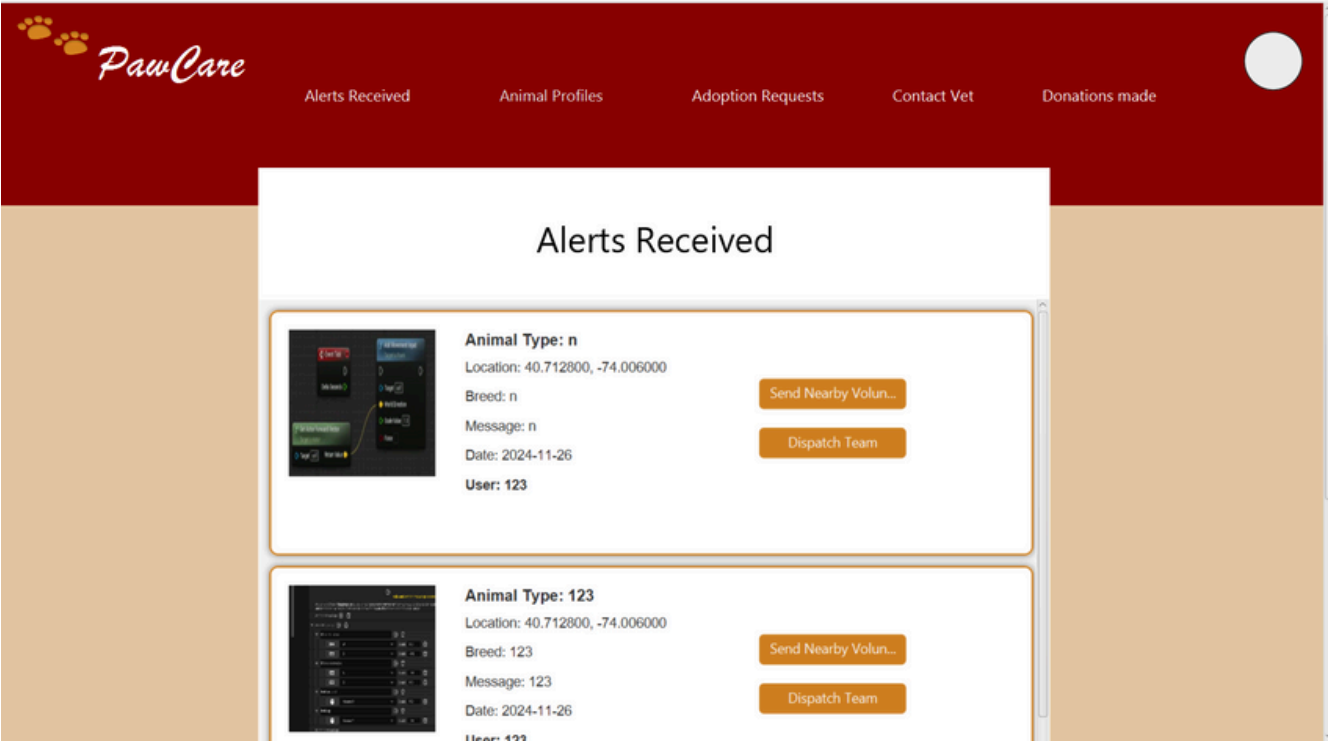
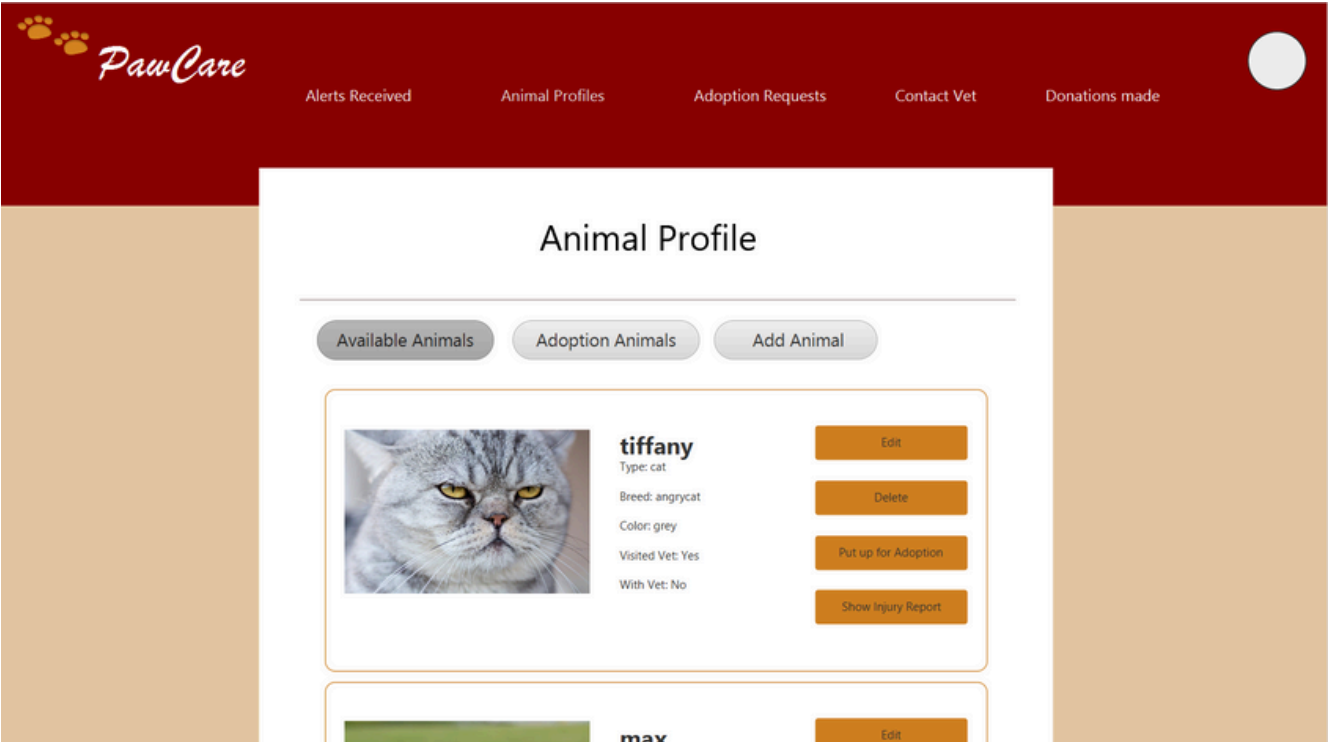
Last Name

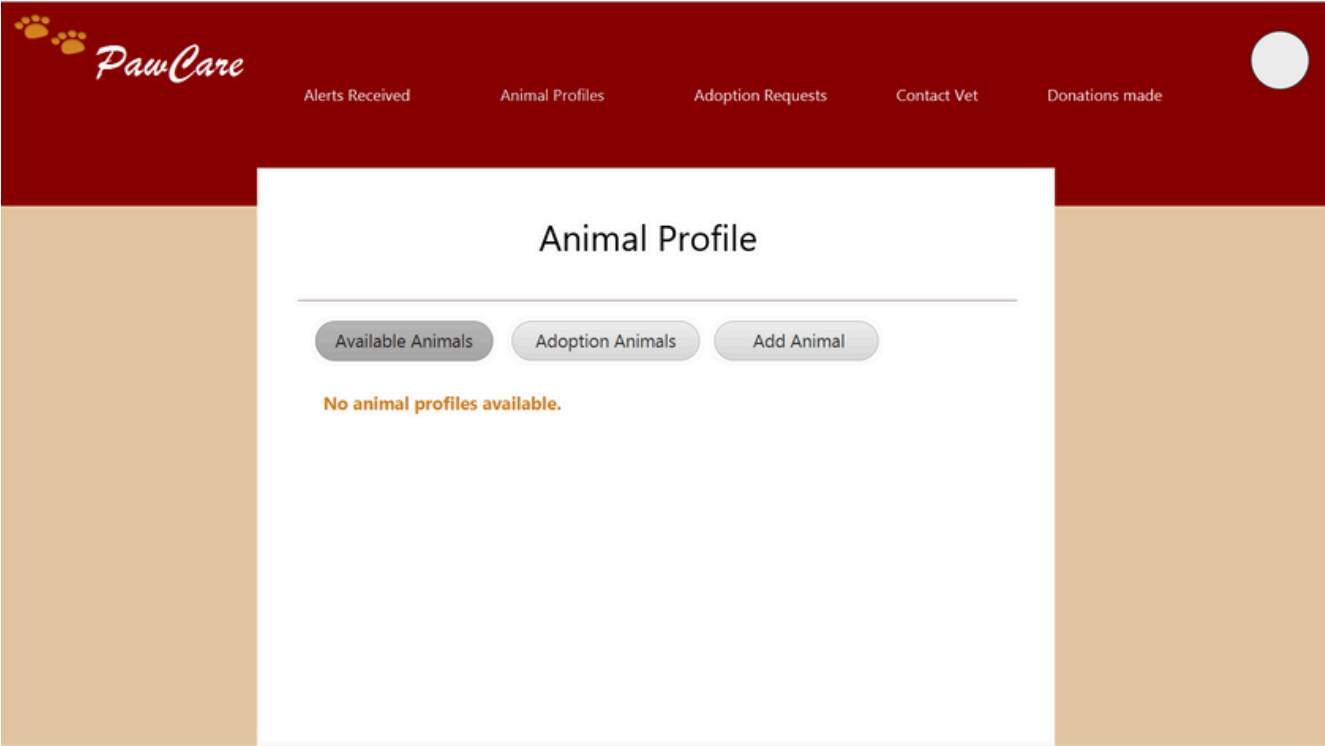
Amount

3. Rescue Center interface:

The following are the different pages for the Rescue Center







Animal Profile

Name:

Type:

Breed:

Color:

Temperature:

Heart Rate:

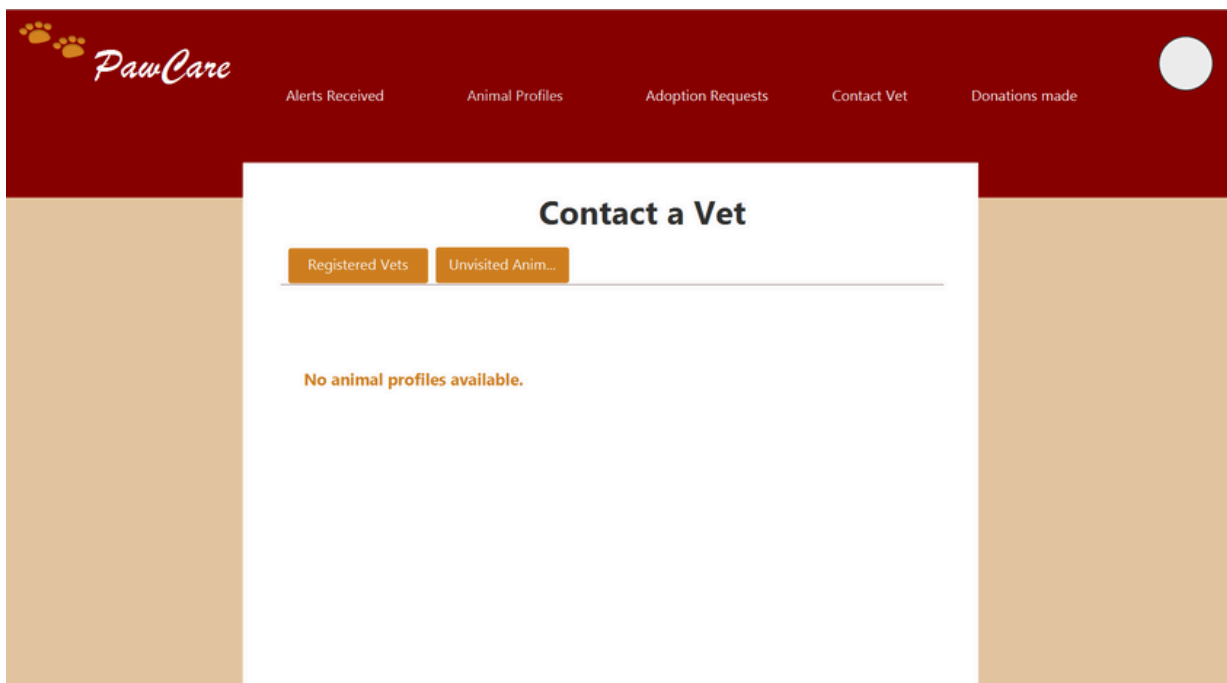
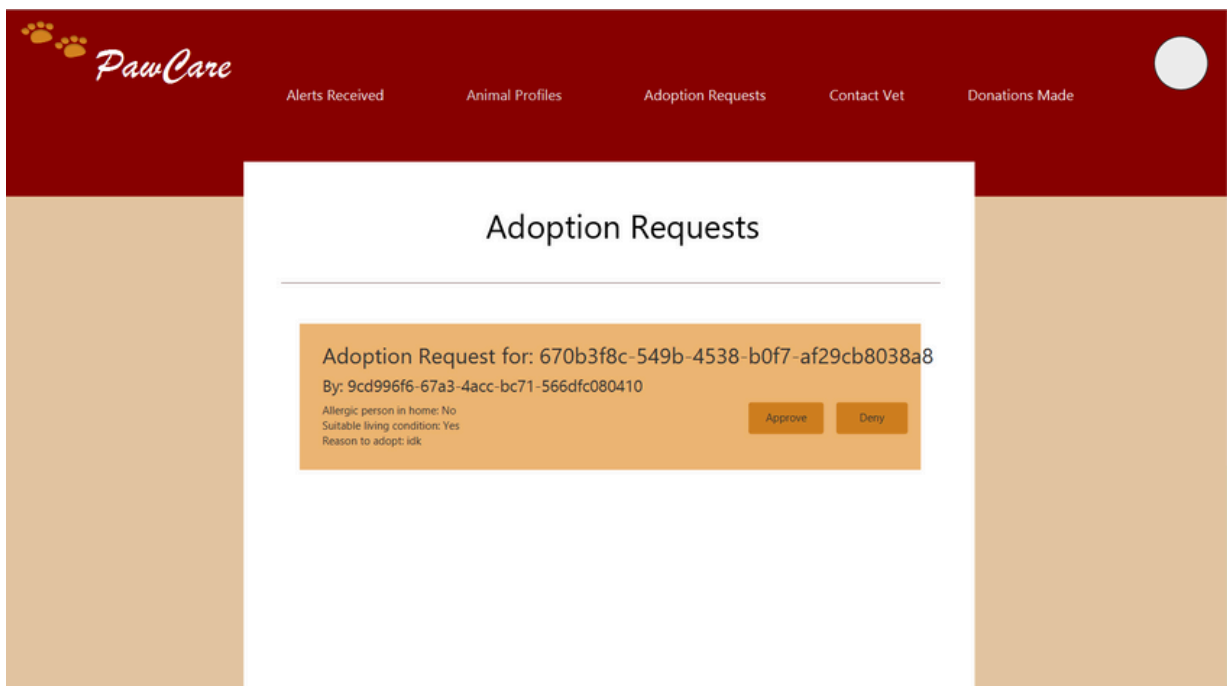
Respiratory Rate:

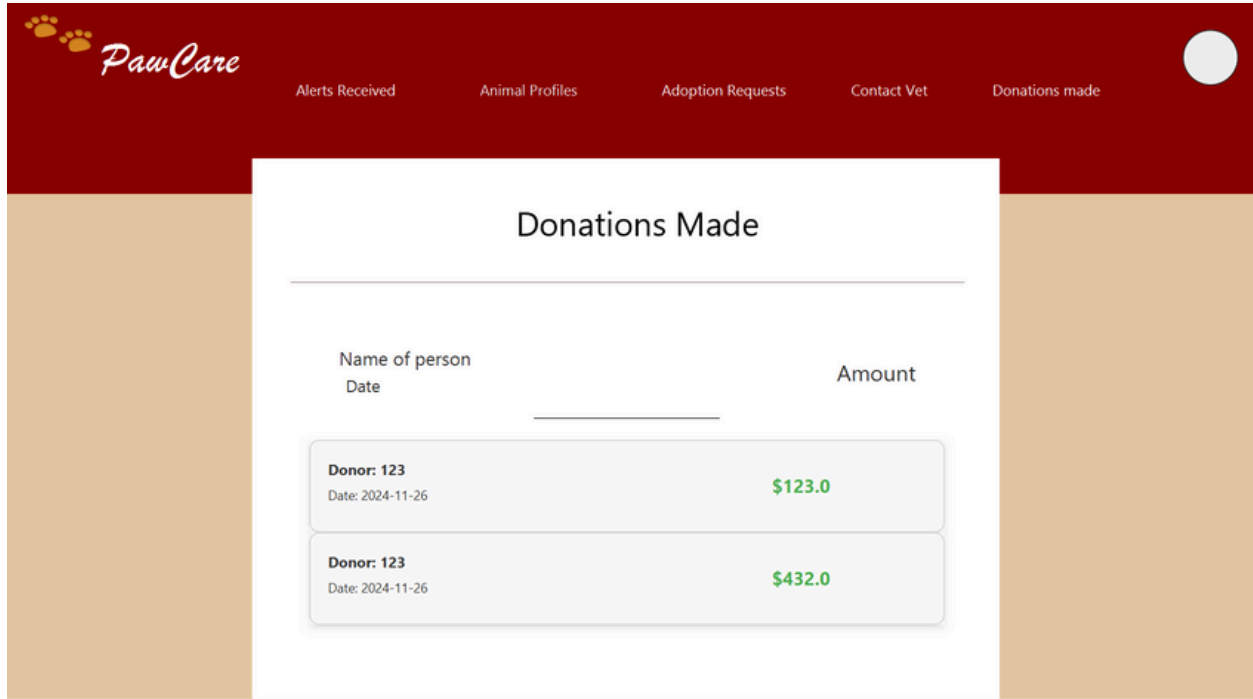
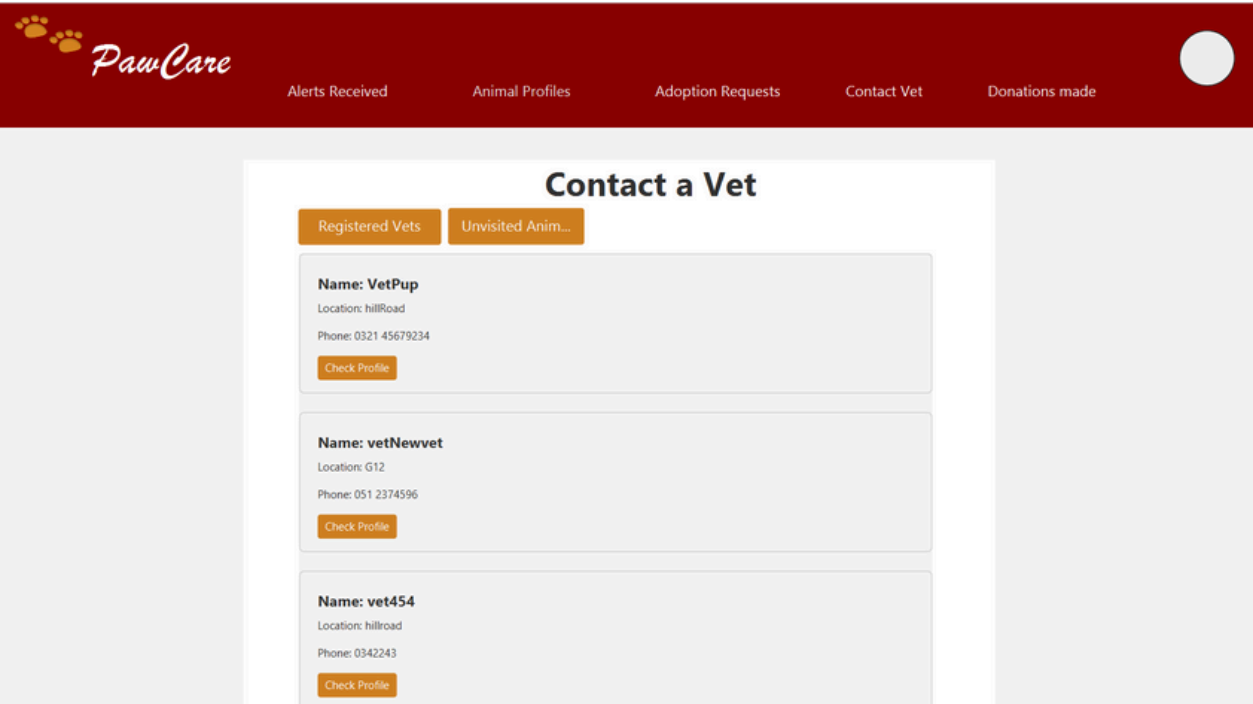
Capillary Refill Time:

Blood Oxygen Level:

Blood Glucose Level:

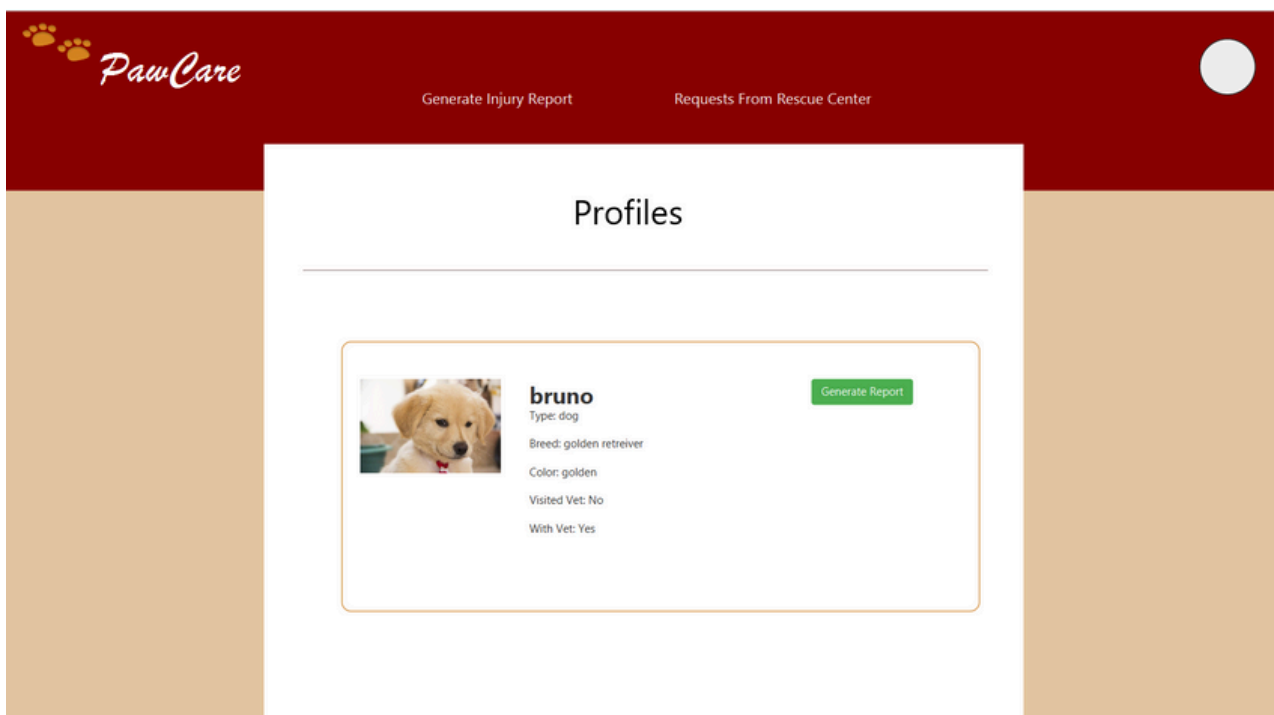
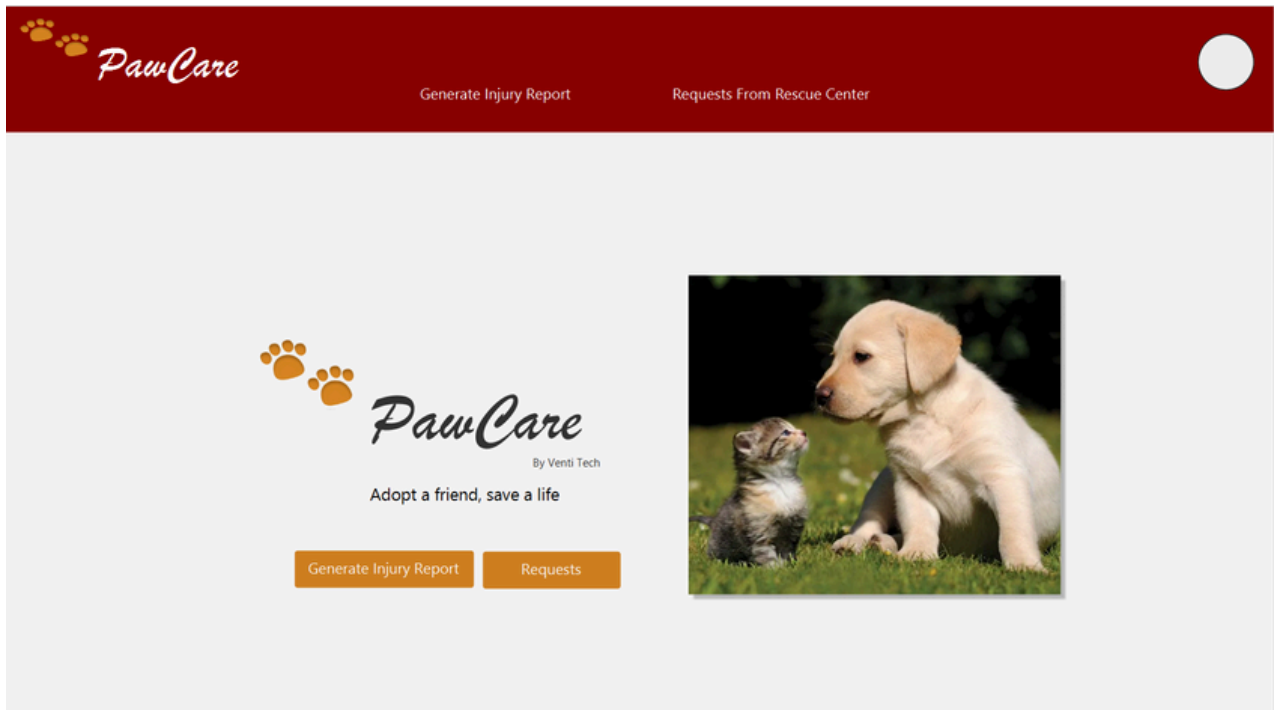
Weight:

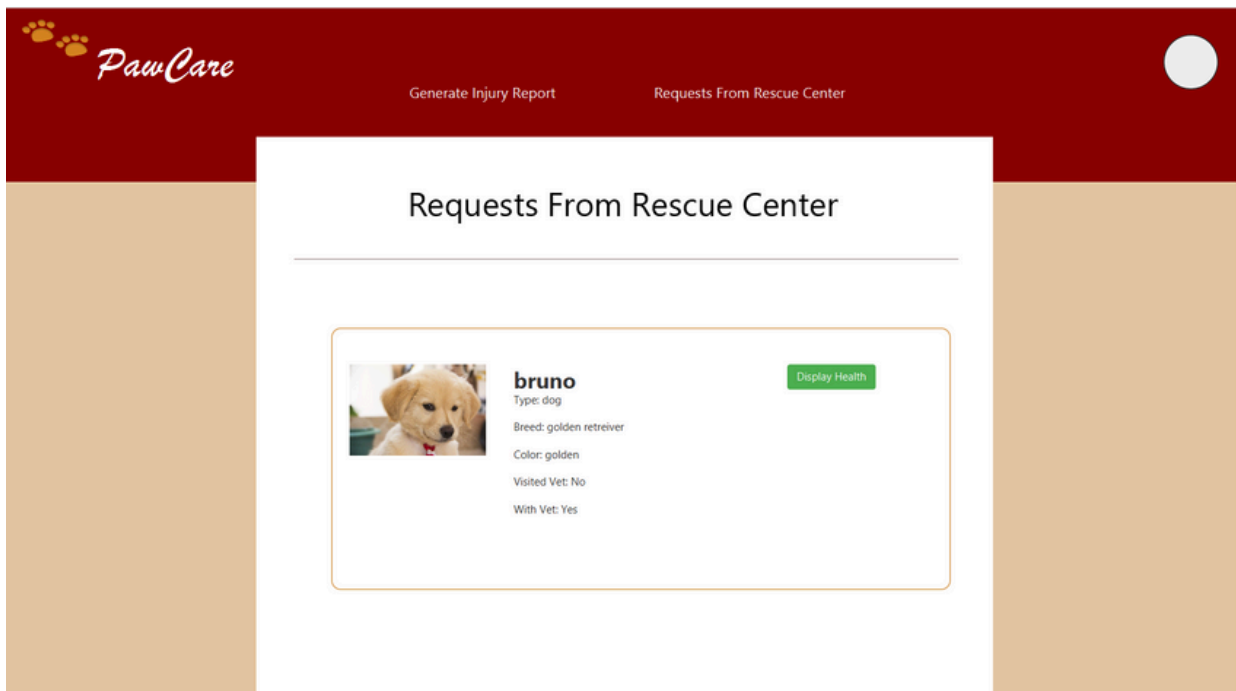




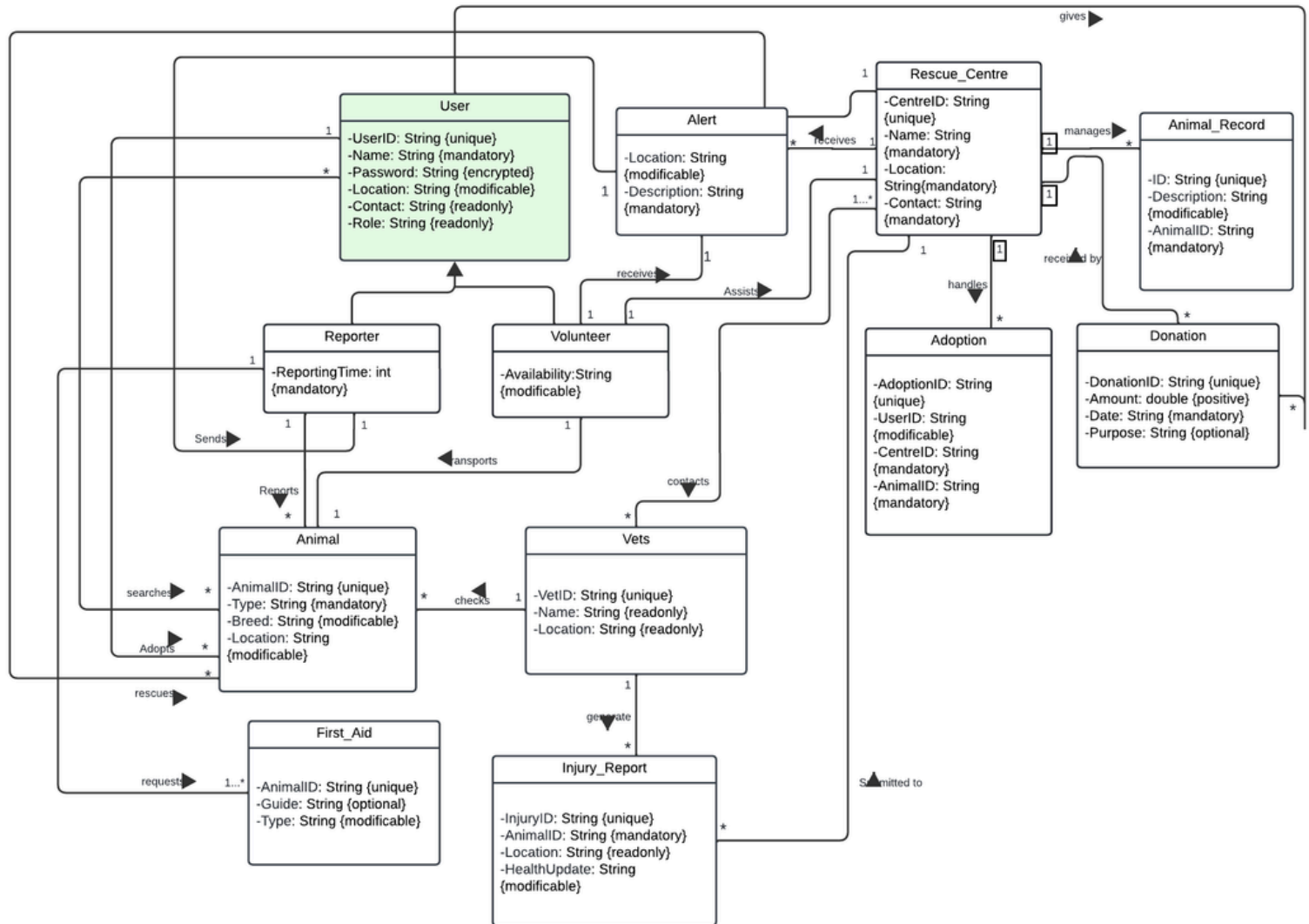
Vets interface:

The following are the different pages for the Vets

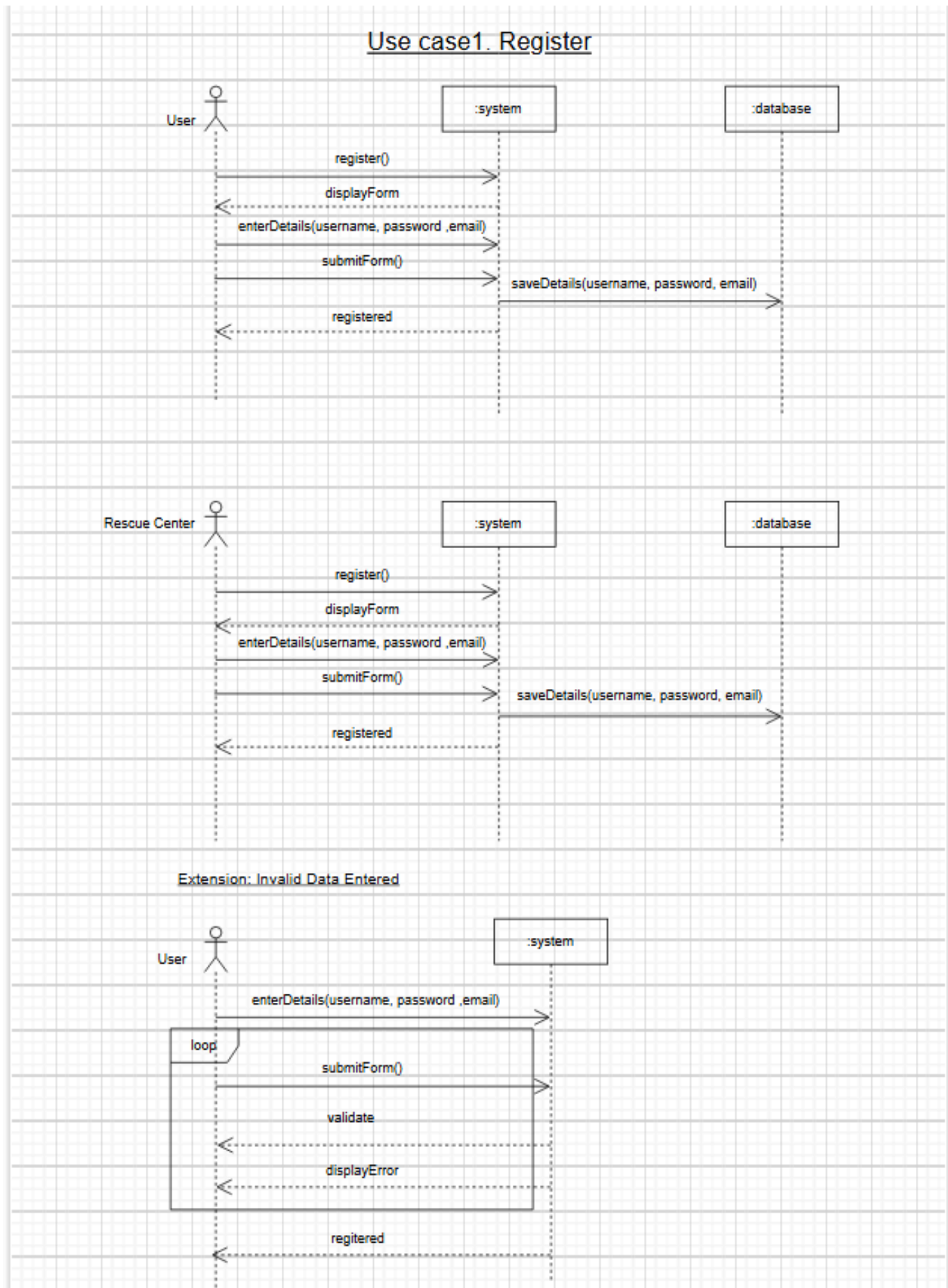


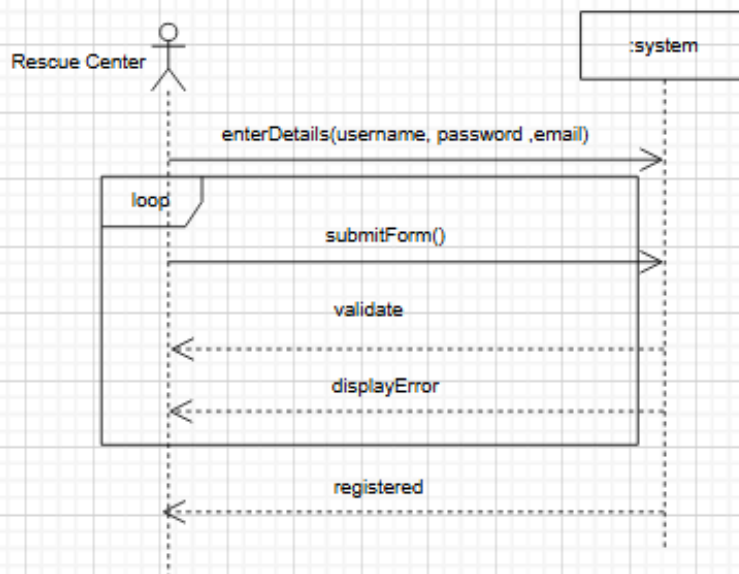


4. Domain Model

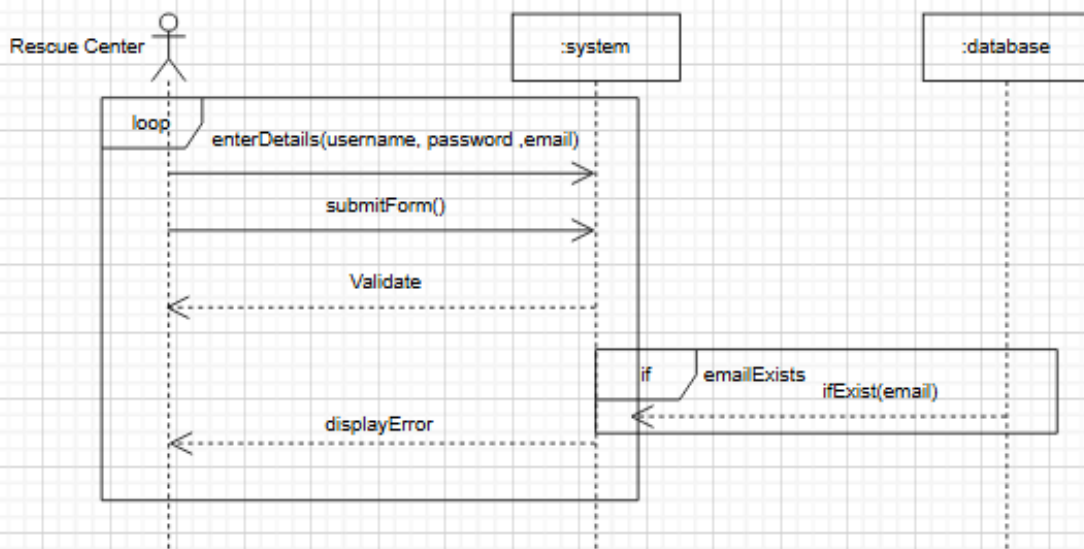
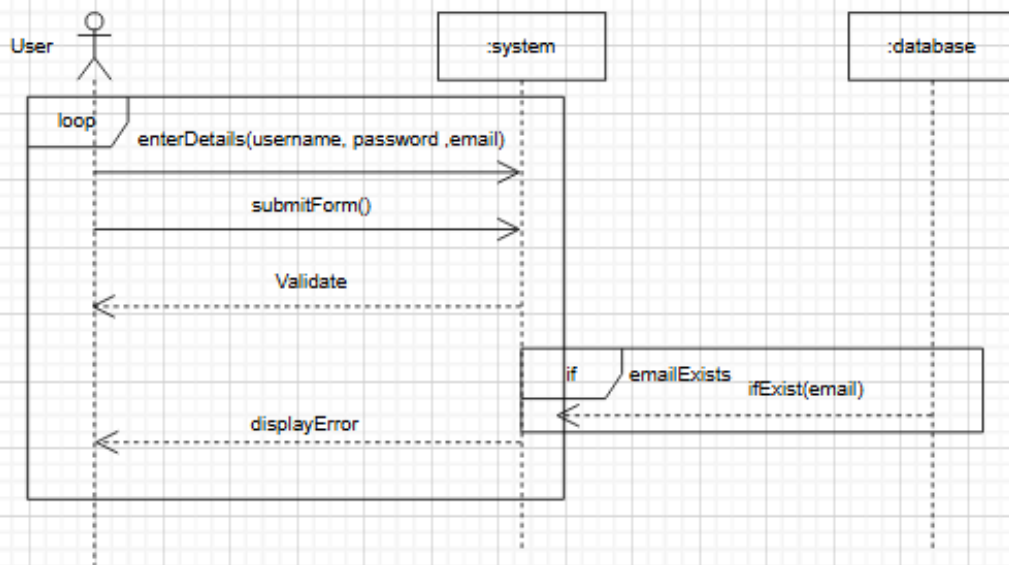


5. System Sequence Diagram

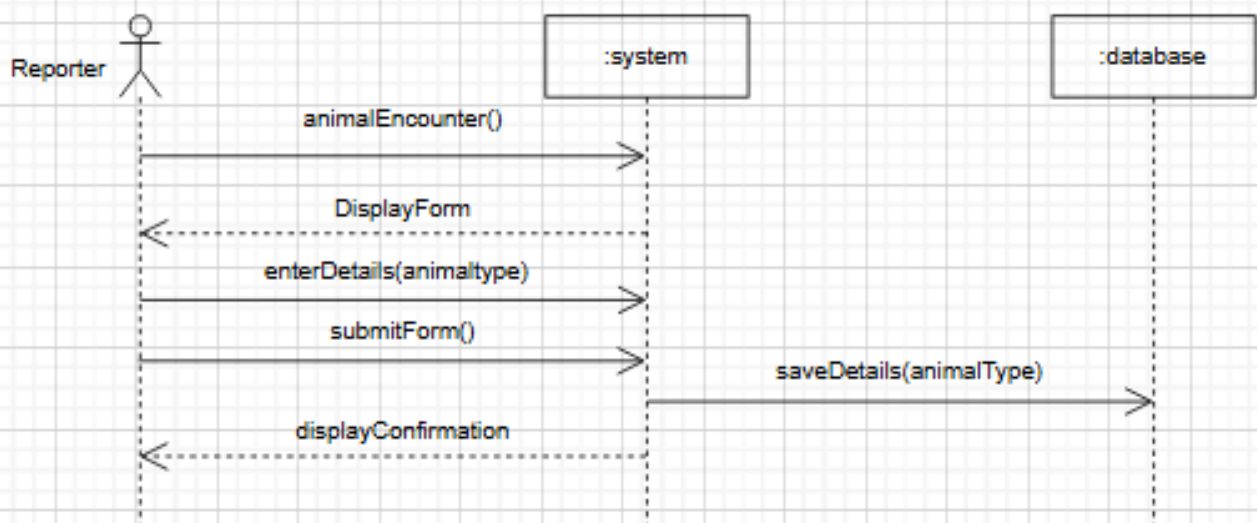




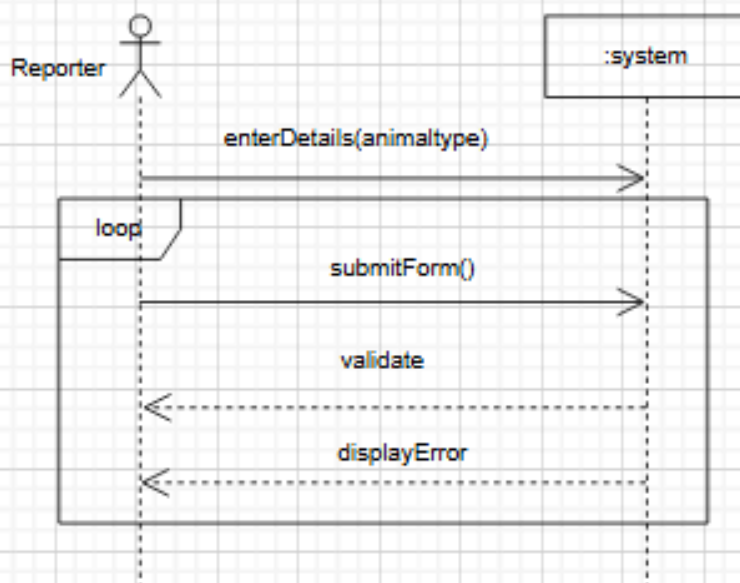
Extension: Already Registered



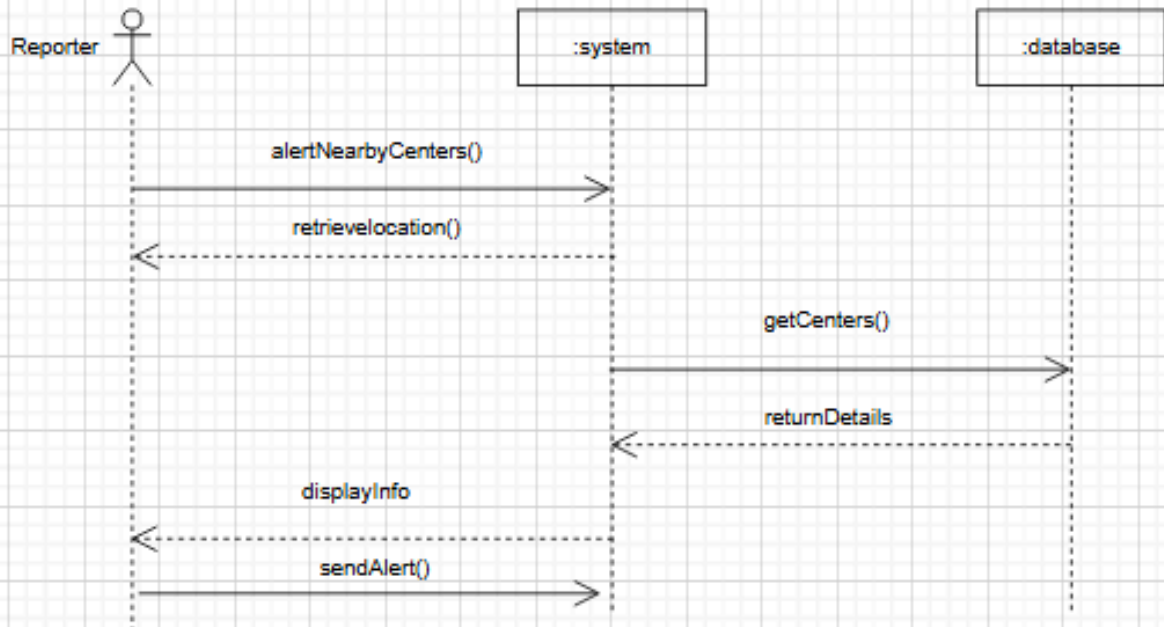
Use case 2: Report Injured Animal



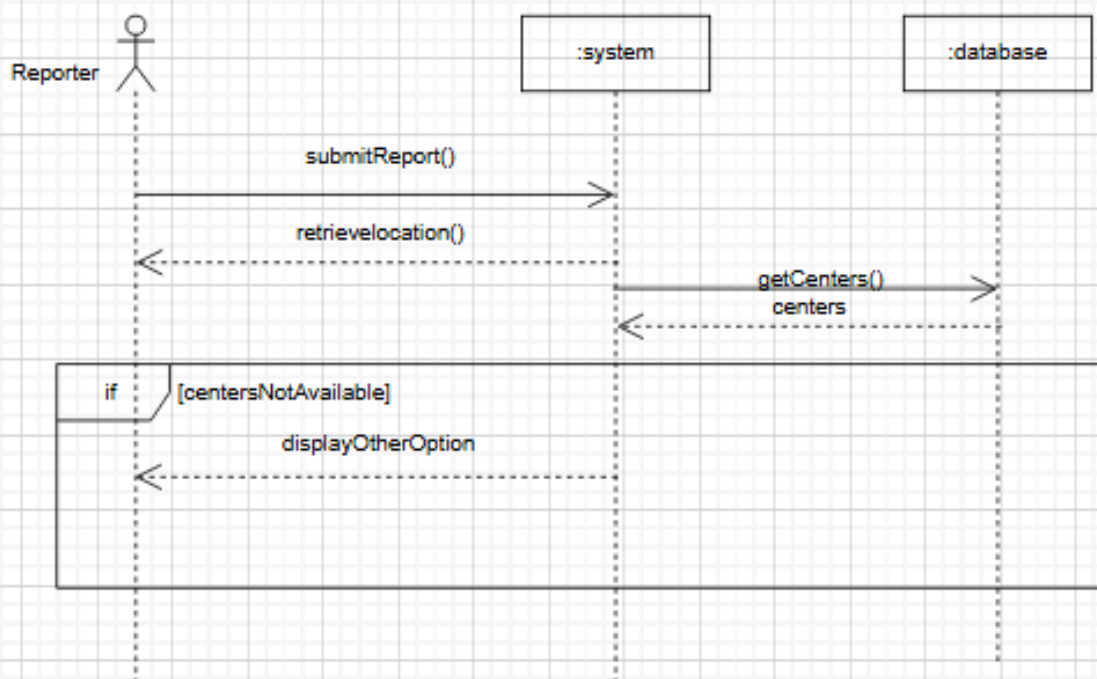
Extension: Incomplete fields



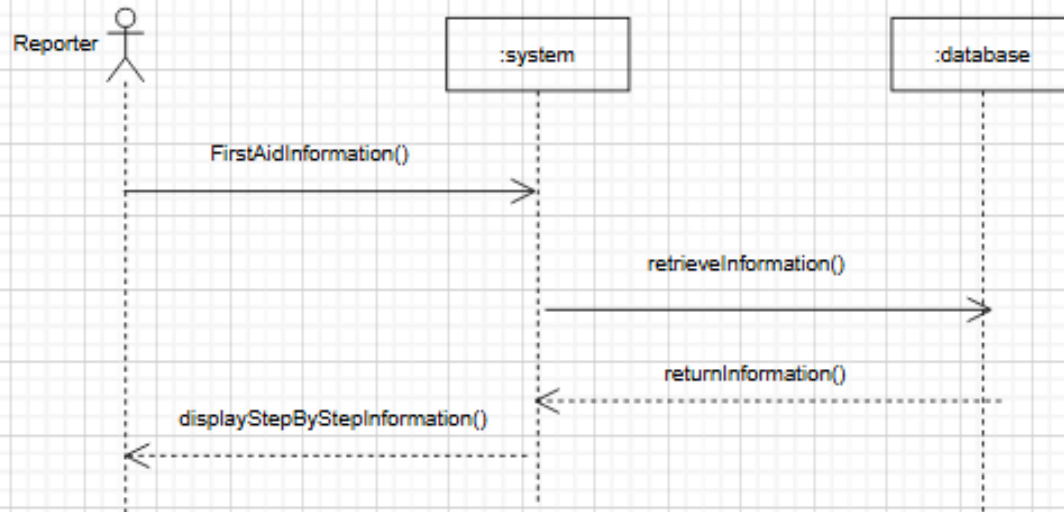
Use case 3: Alert Nearby Centers



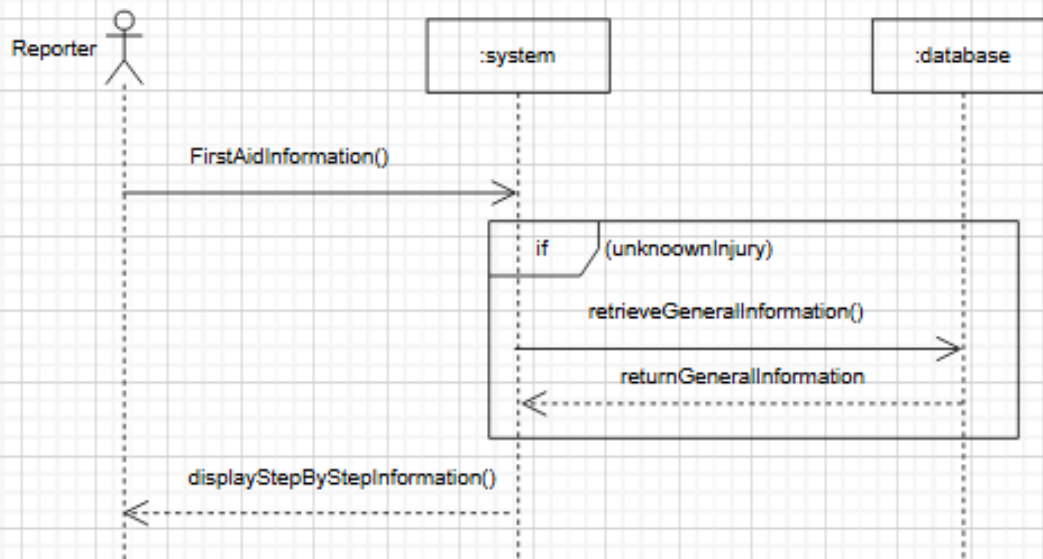
Extension: No centre nearby

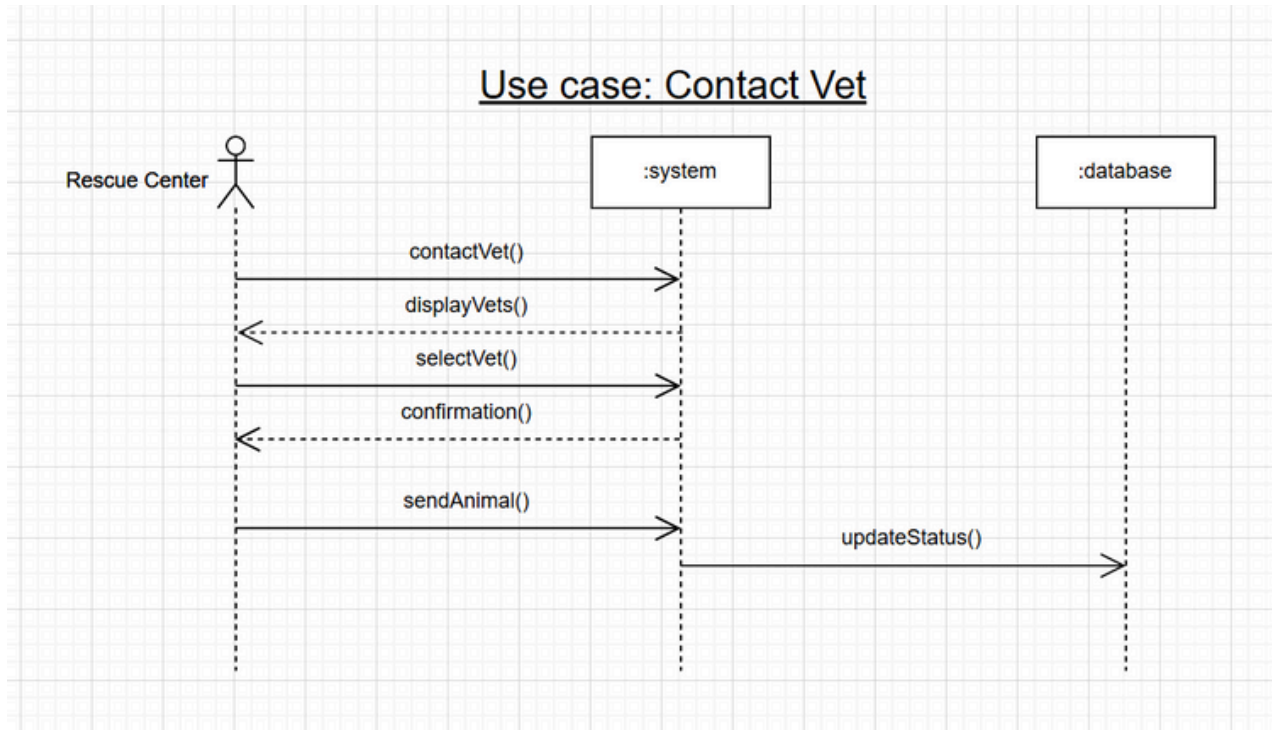


Use case 4: Provide First Aid

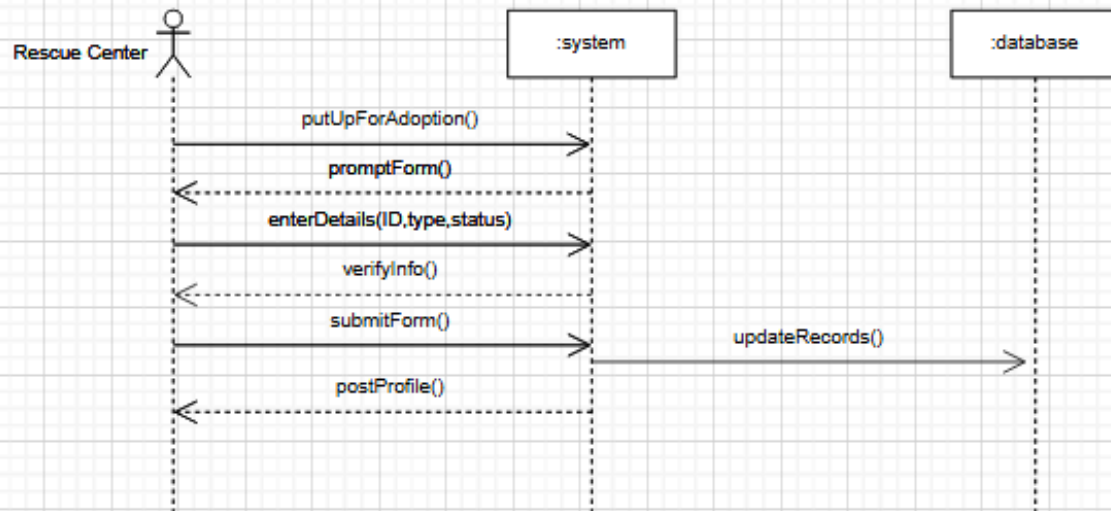


Extension: In case of unknown injury

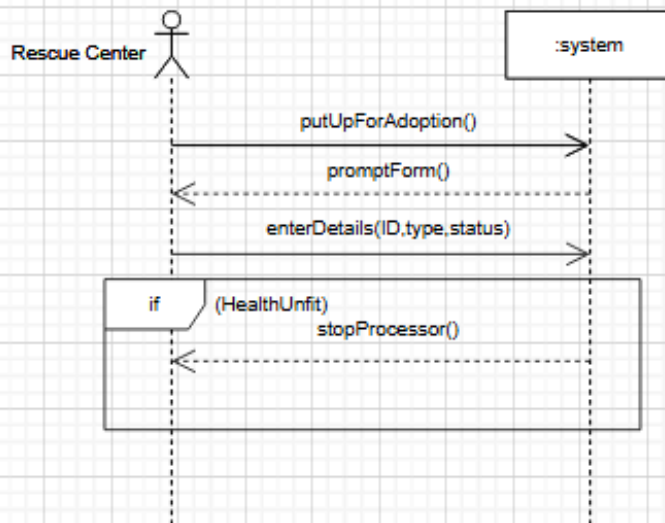




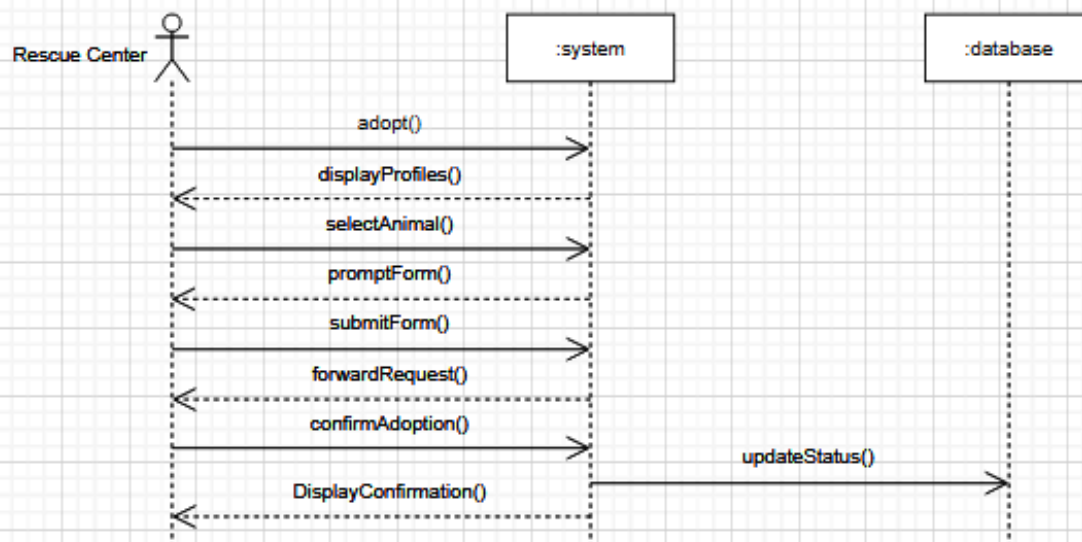
Use case Put Animal up for adoption



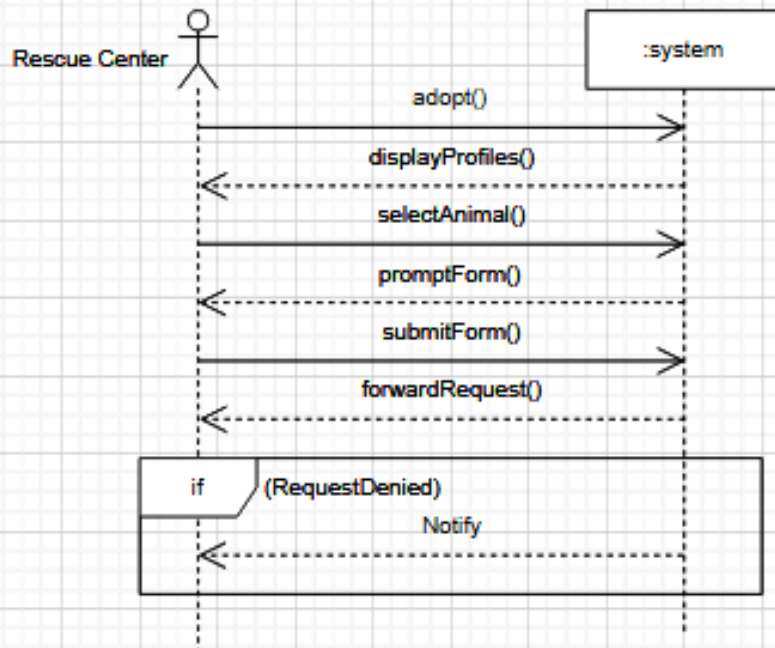
Extension: Animal Died



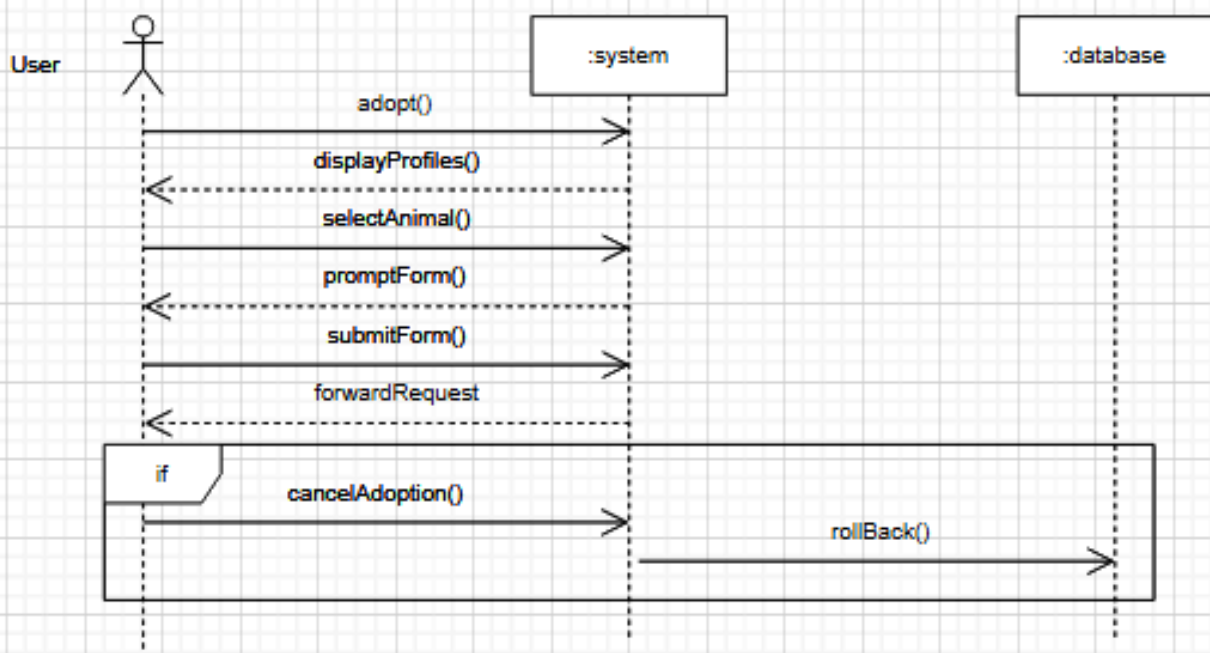
Use case: Adopt Animal



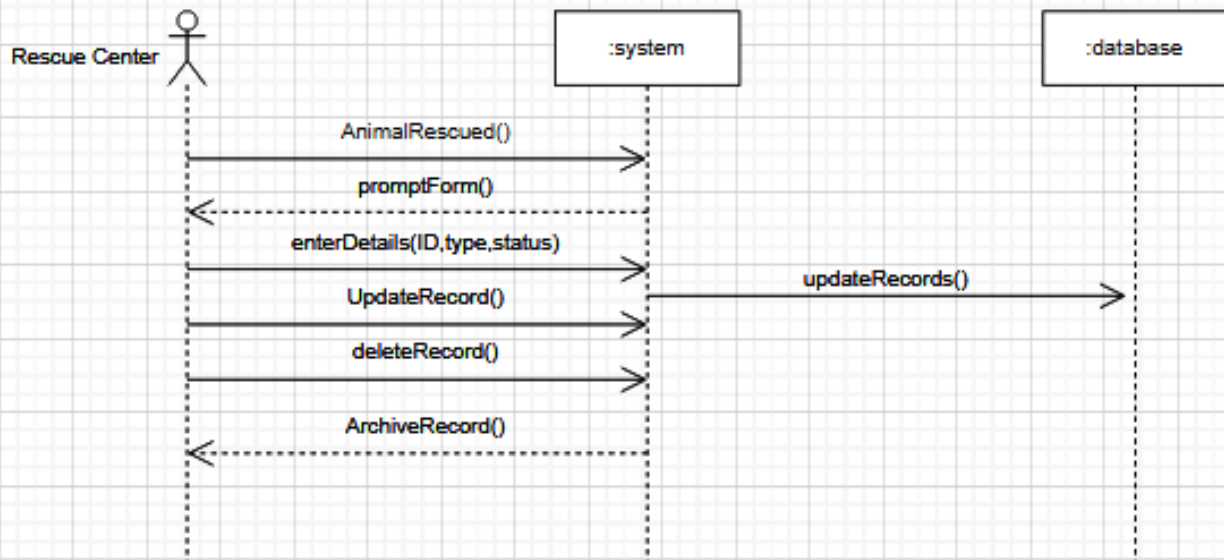
Extension: Application Denied



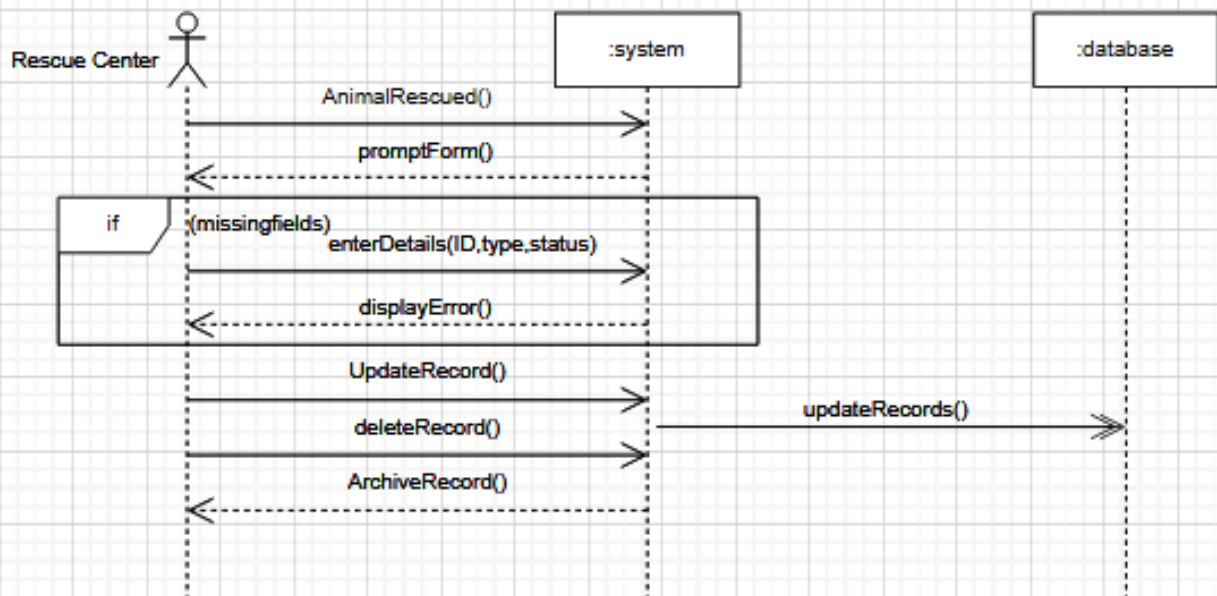
Extension: Cancel Adoption



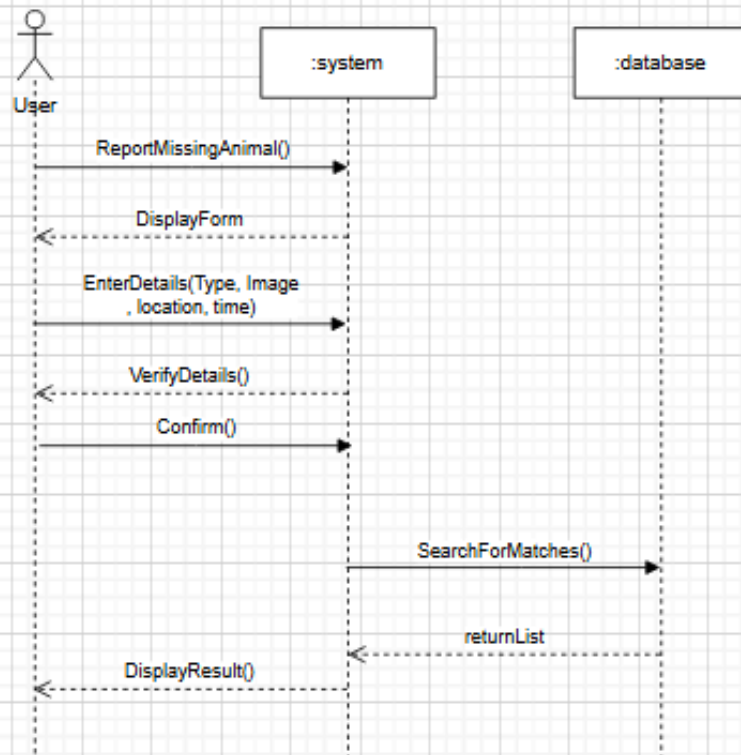
Use case: Manage Animal Records



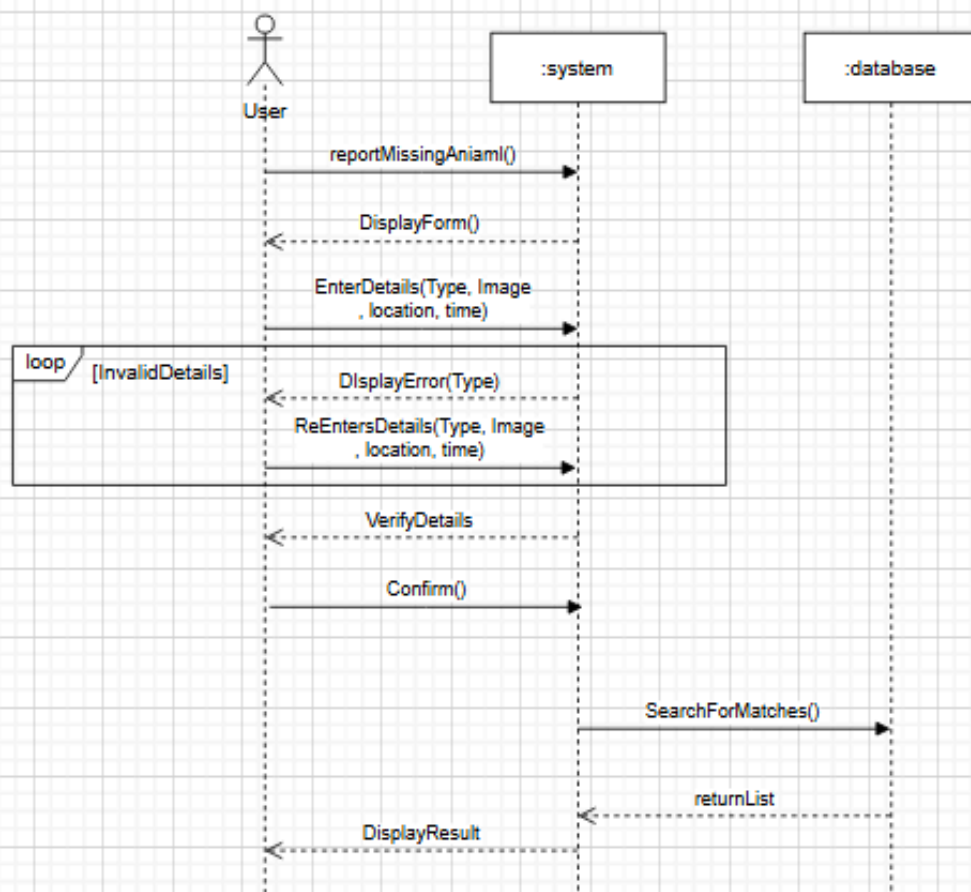
Extension: Missing information



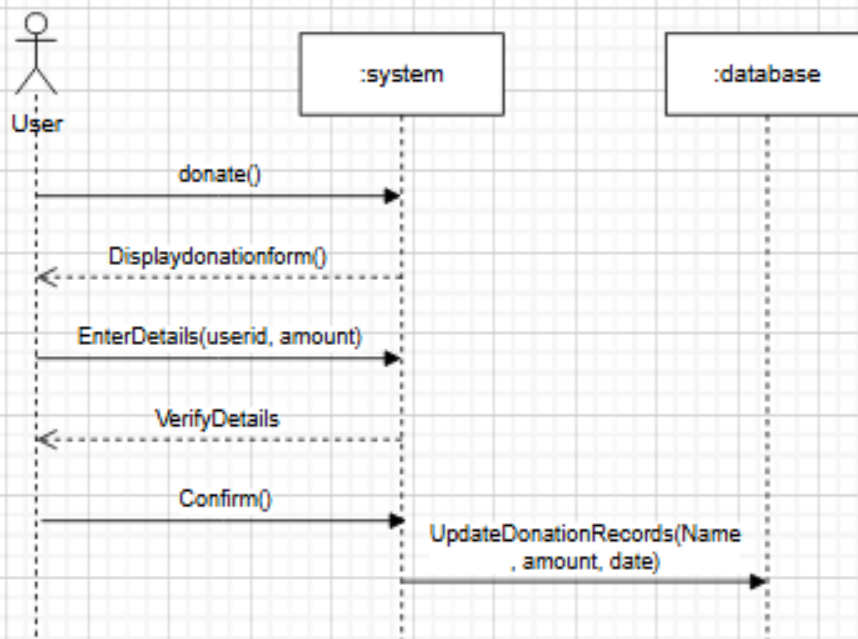
Report Missing Animal scenario



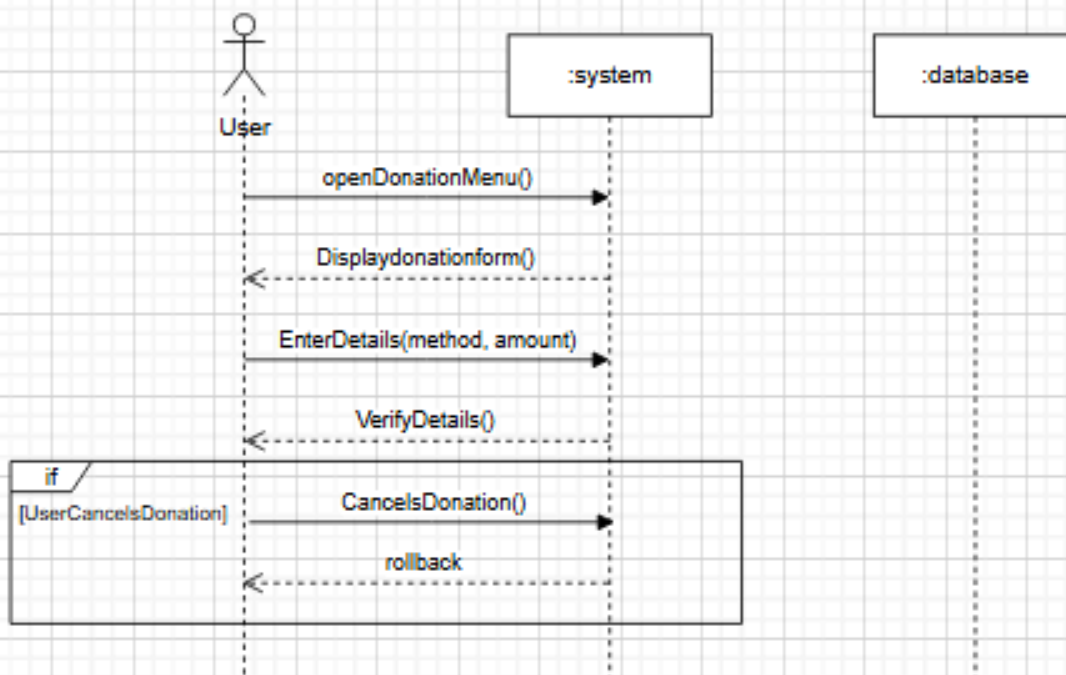
Extension: Invalid/Incomplete Information



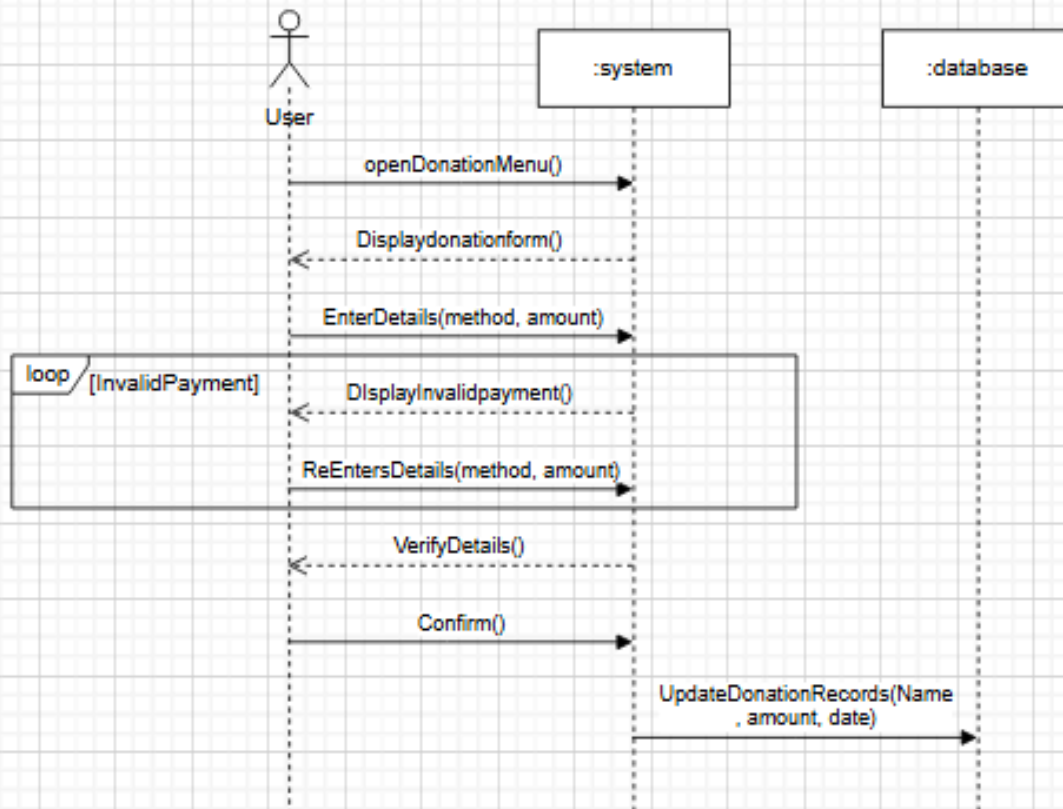
Donate scenario



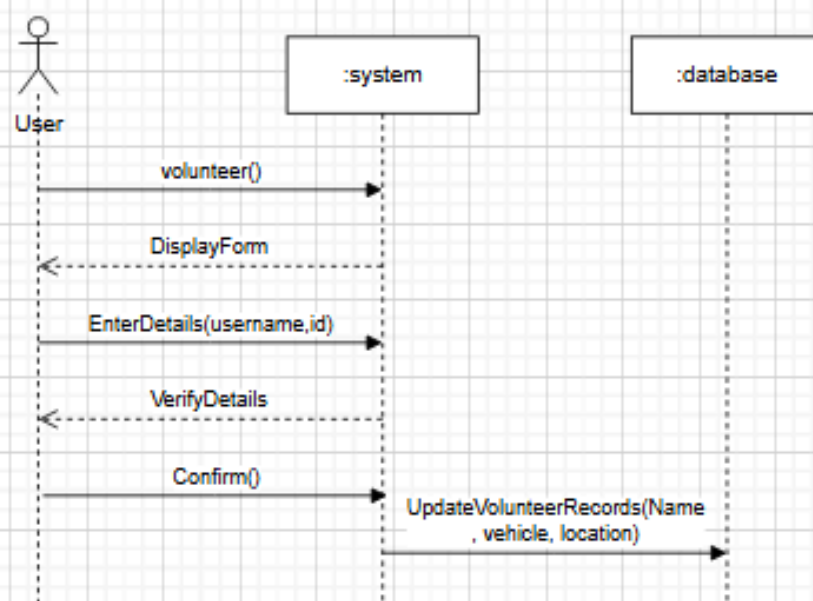
Extension: Cancels Donation



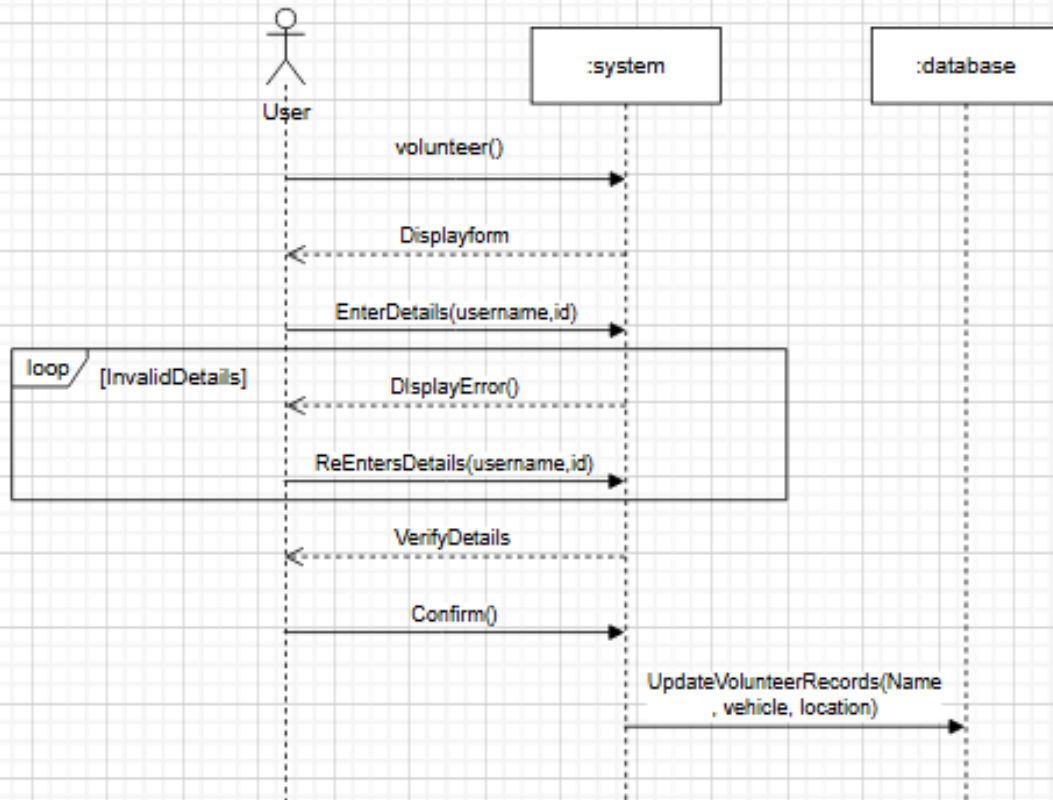
Extension: Invalid Payment



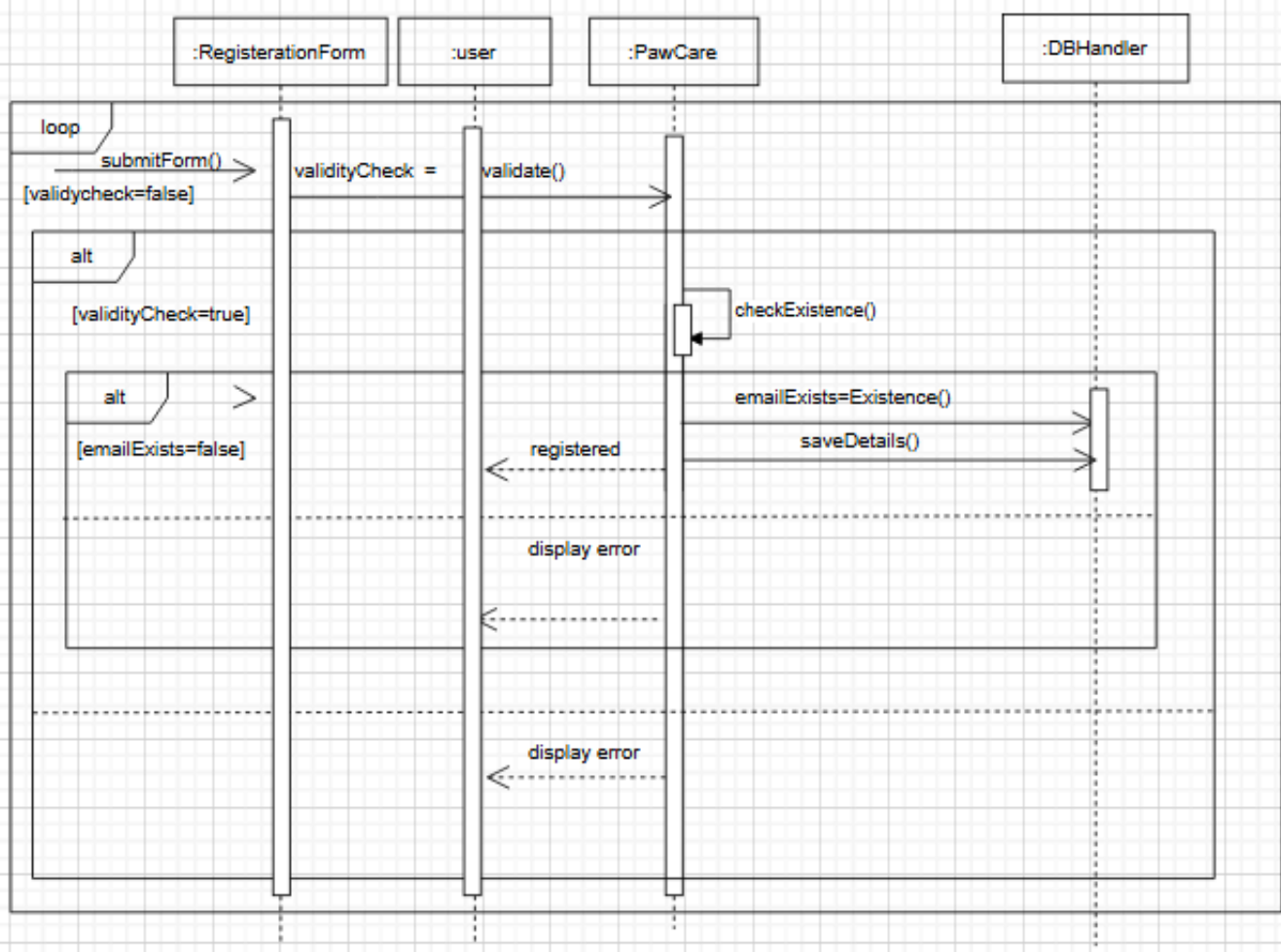
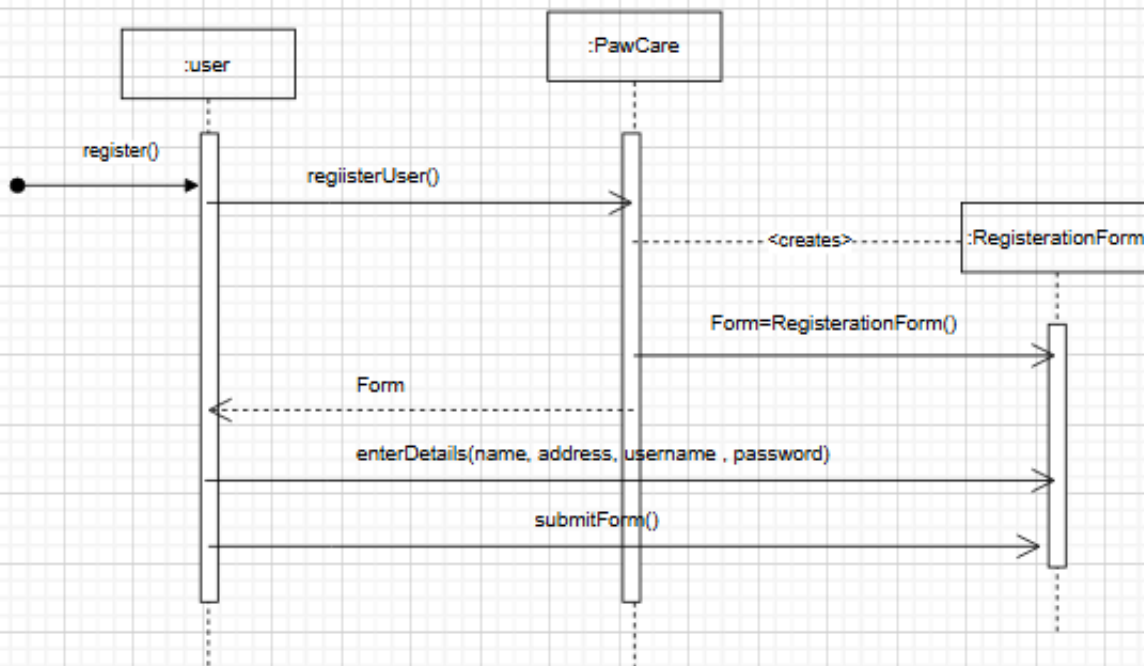
Volunteer scenario

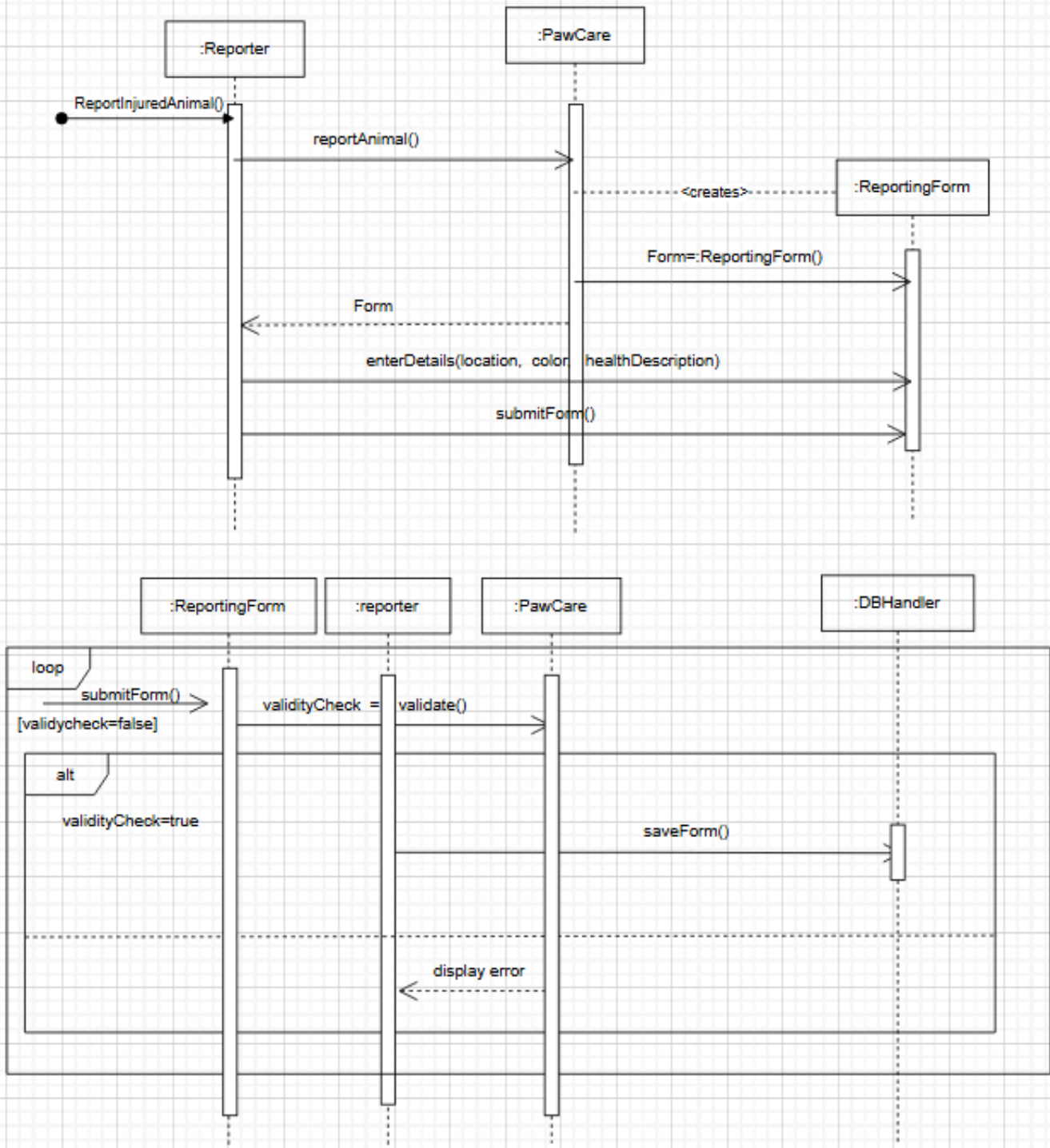


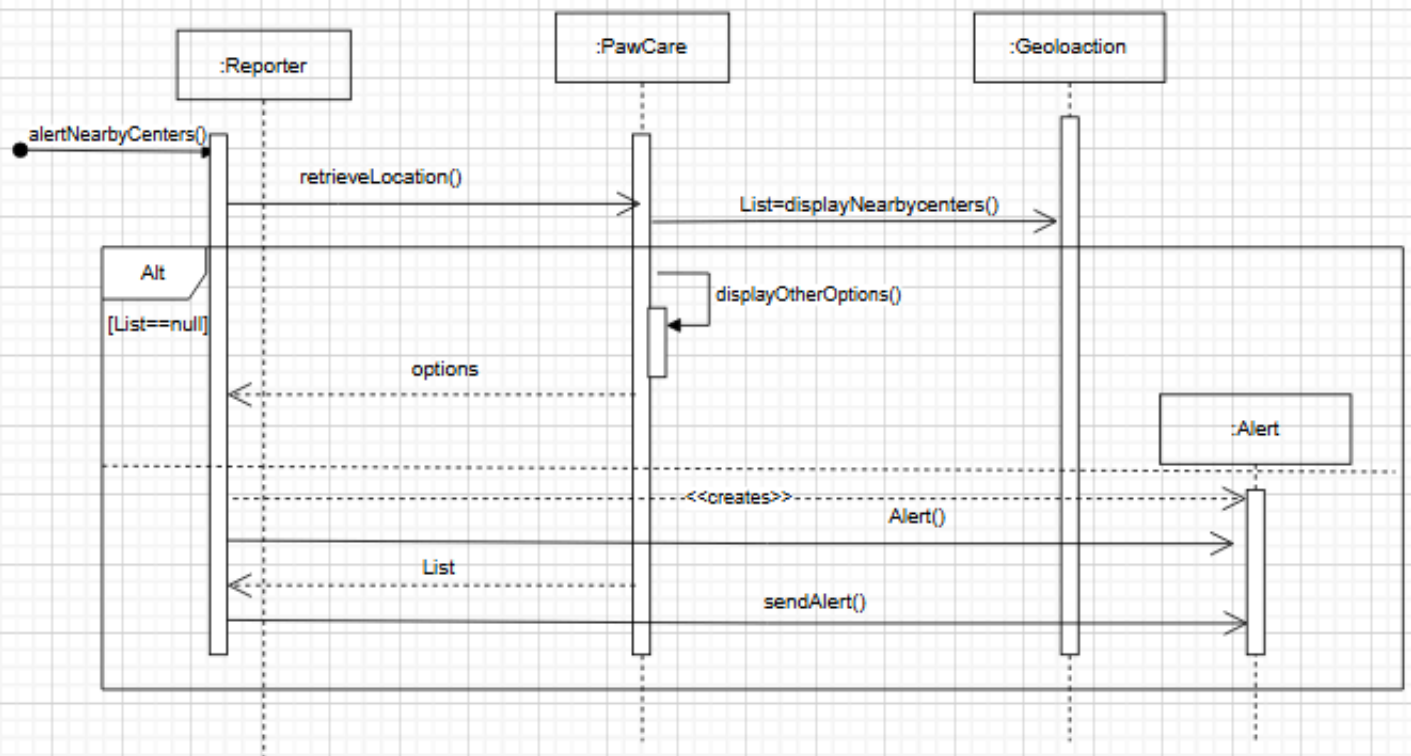
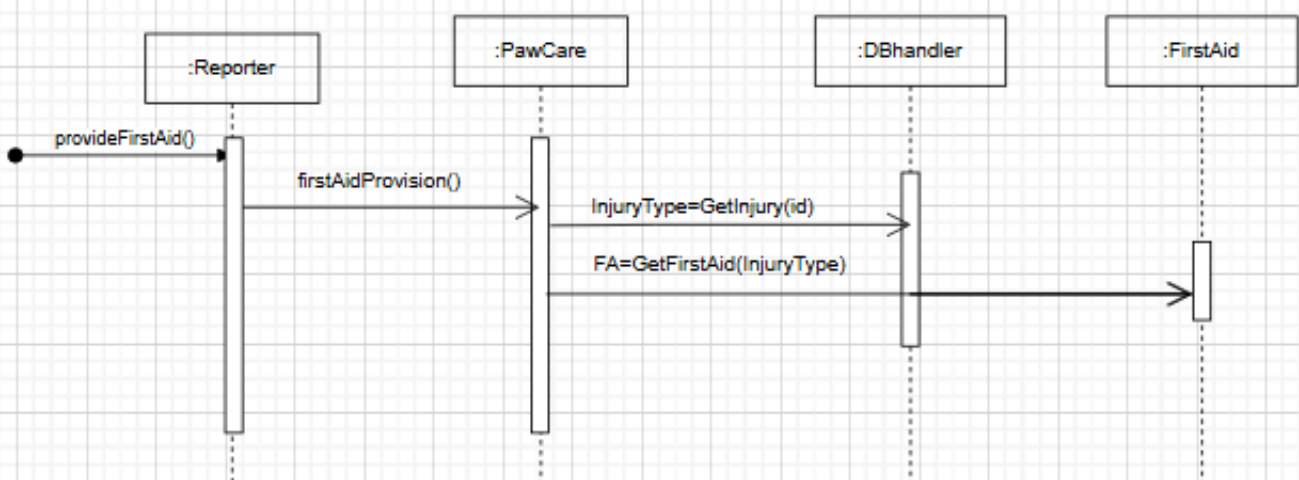
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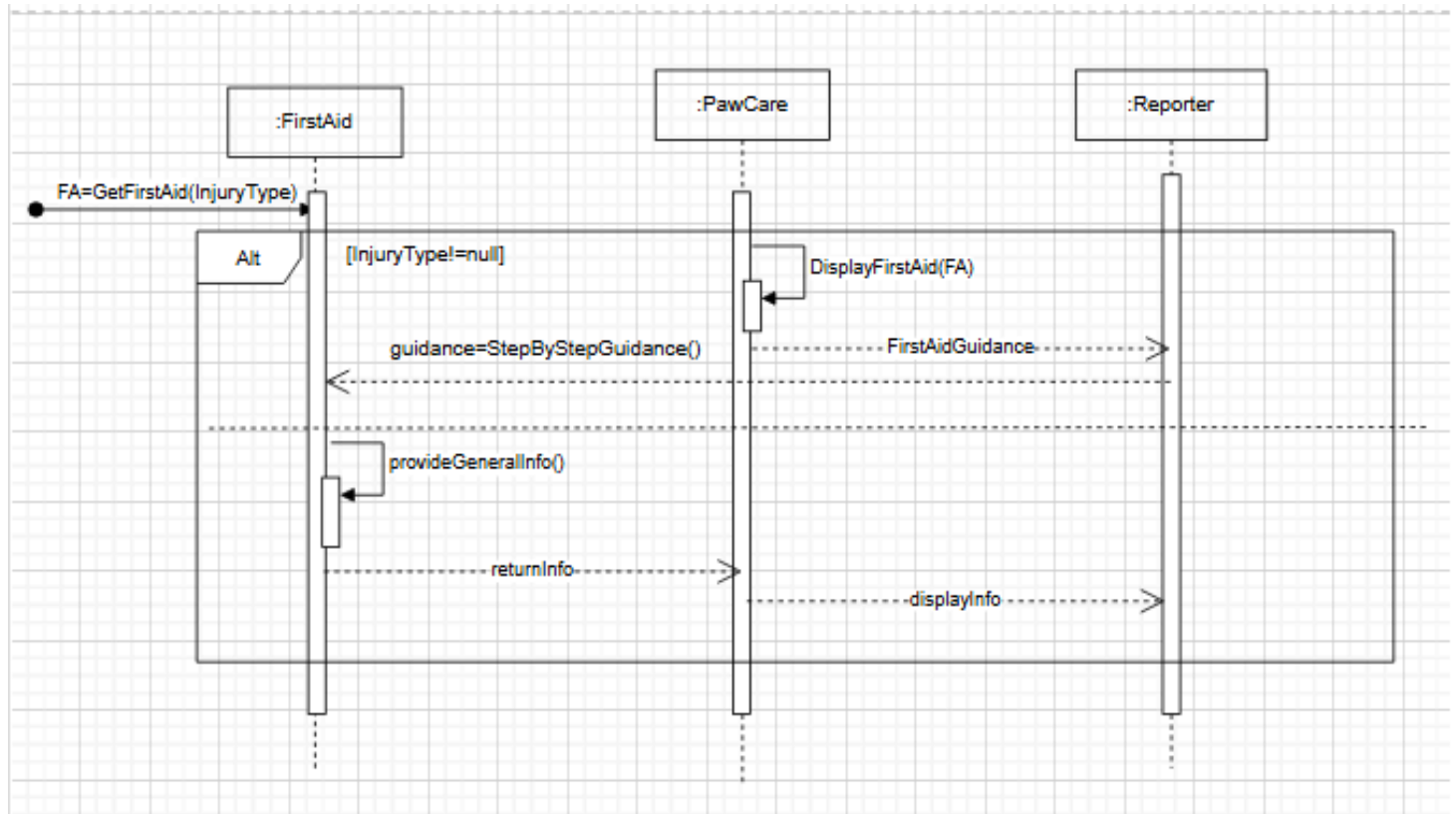


6. Sequence Diagram

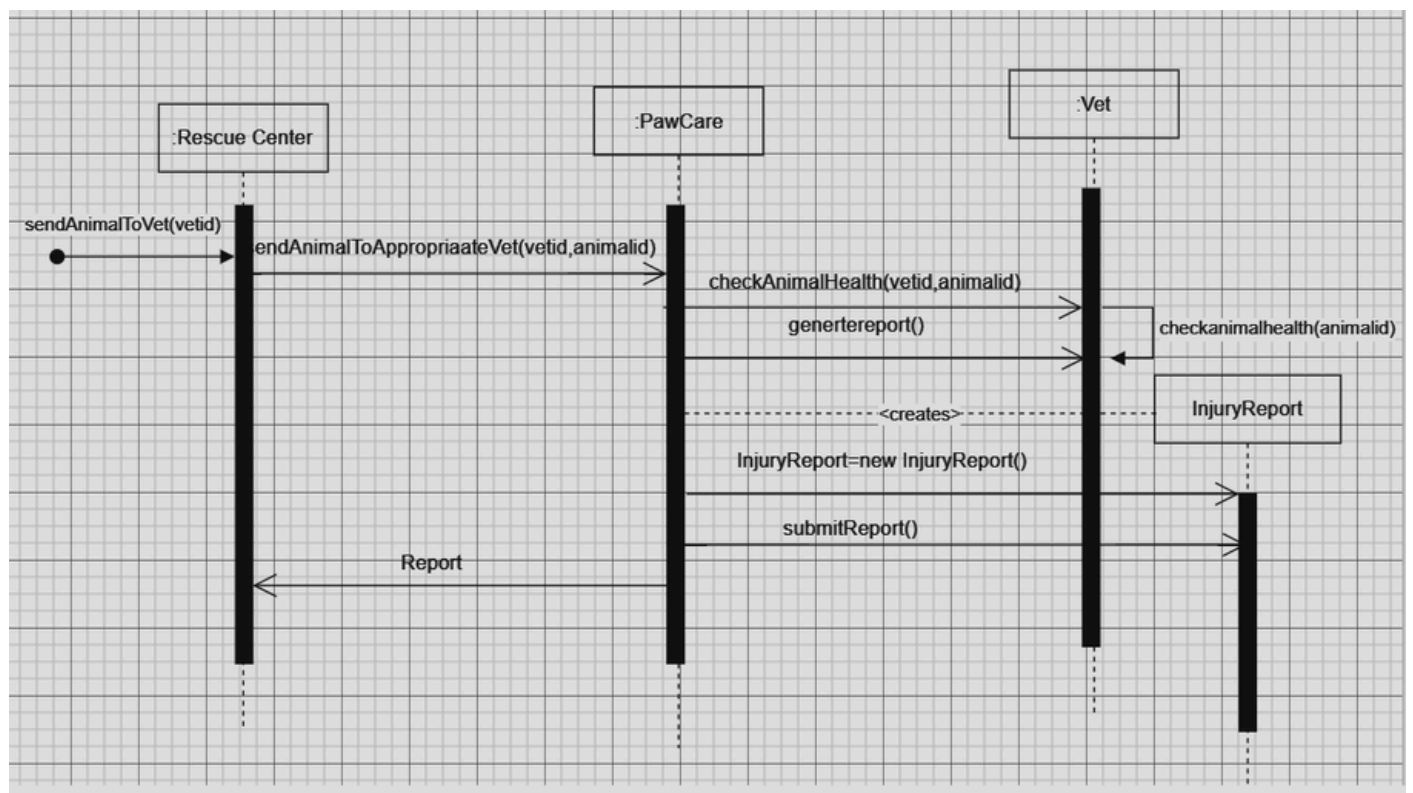


Use case: Report Injured Animal

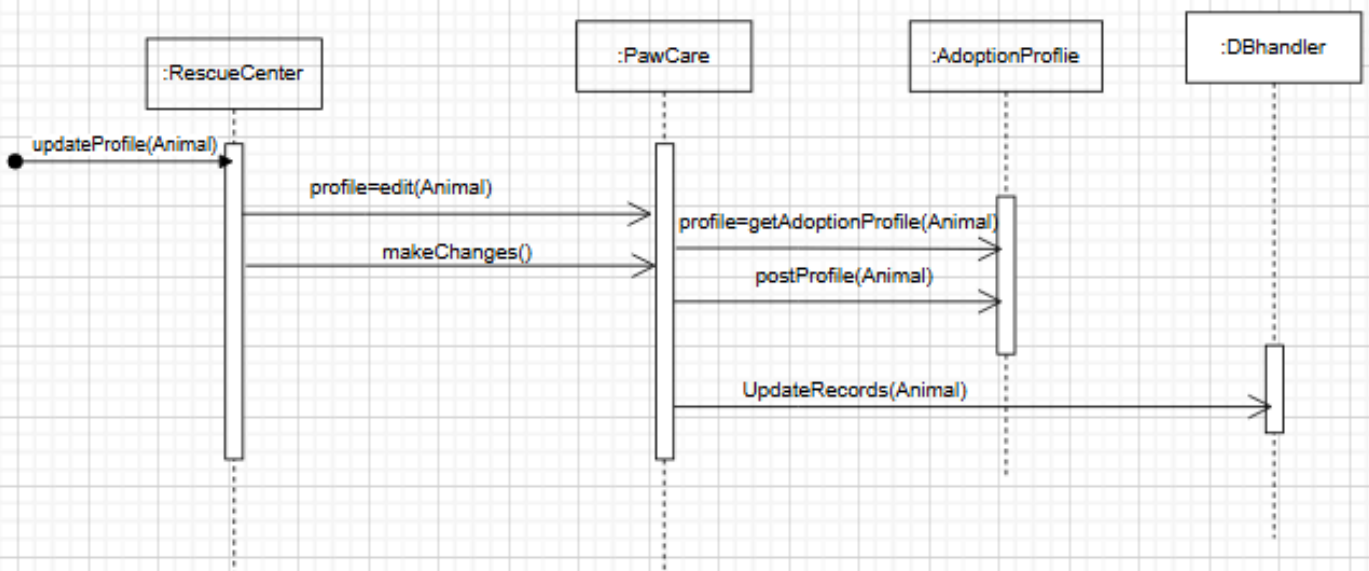
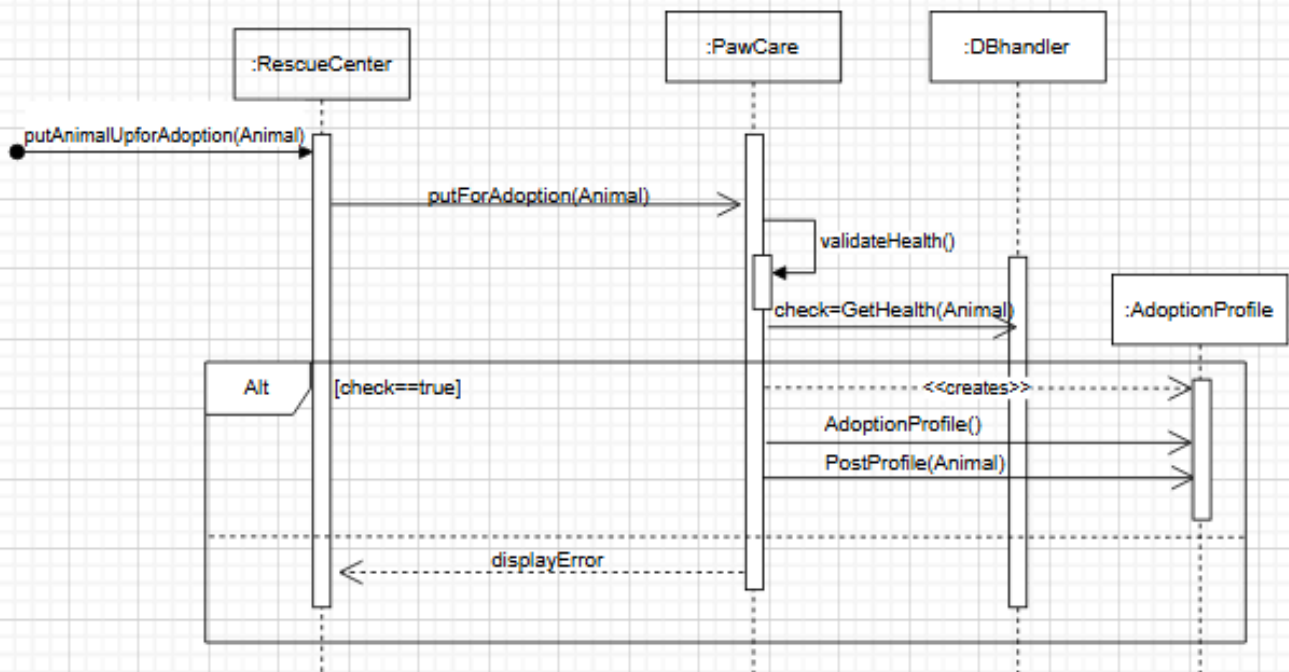
Use case: Alert nearby centerUse case: Provide first aid

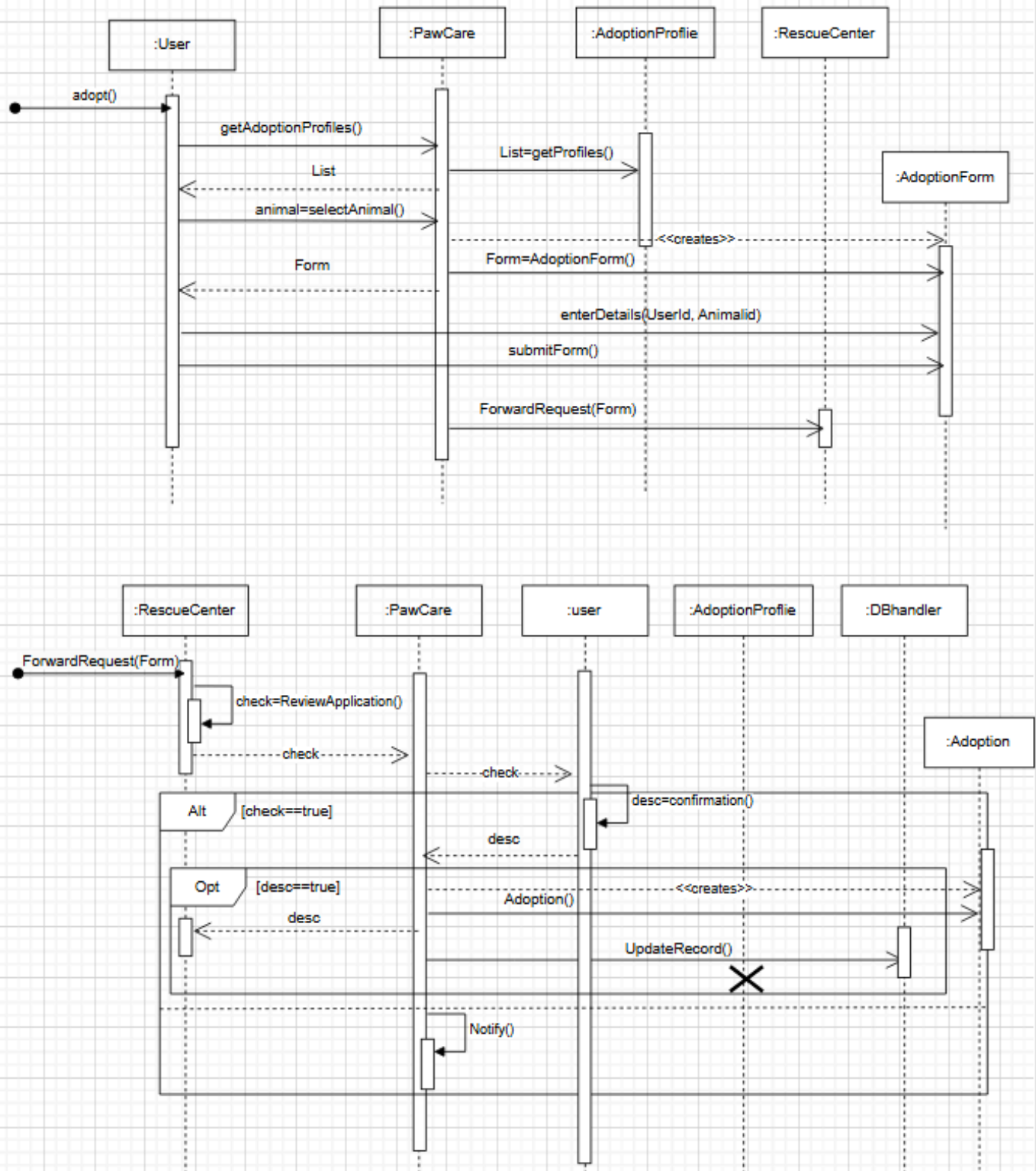


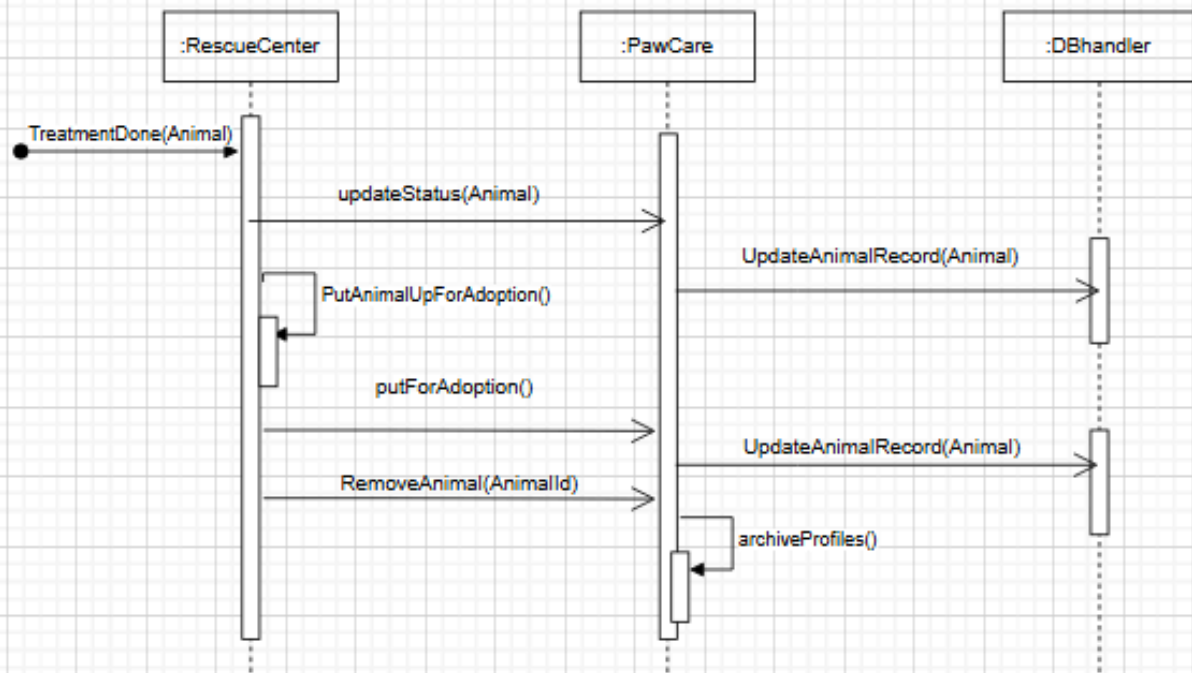
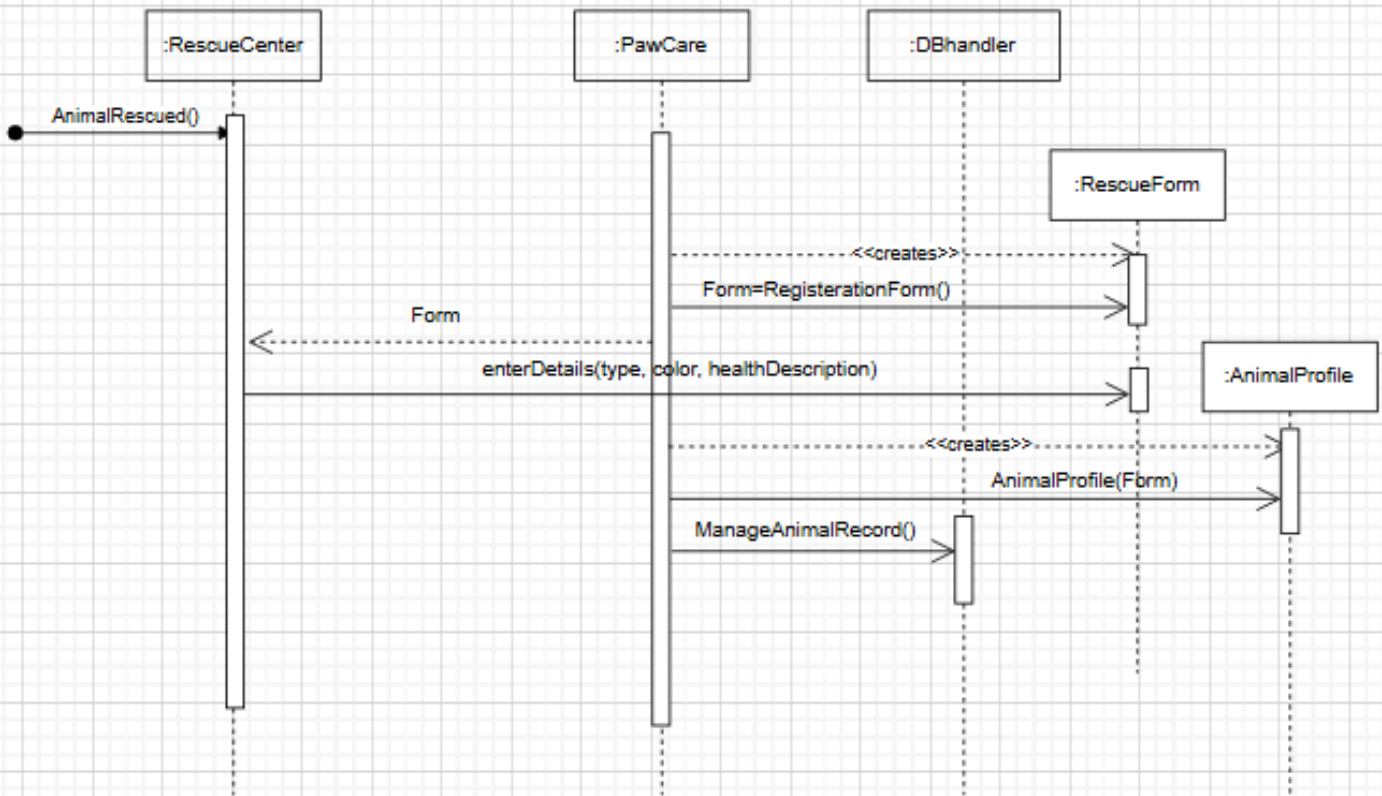
Usecase: Contact vet

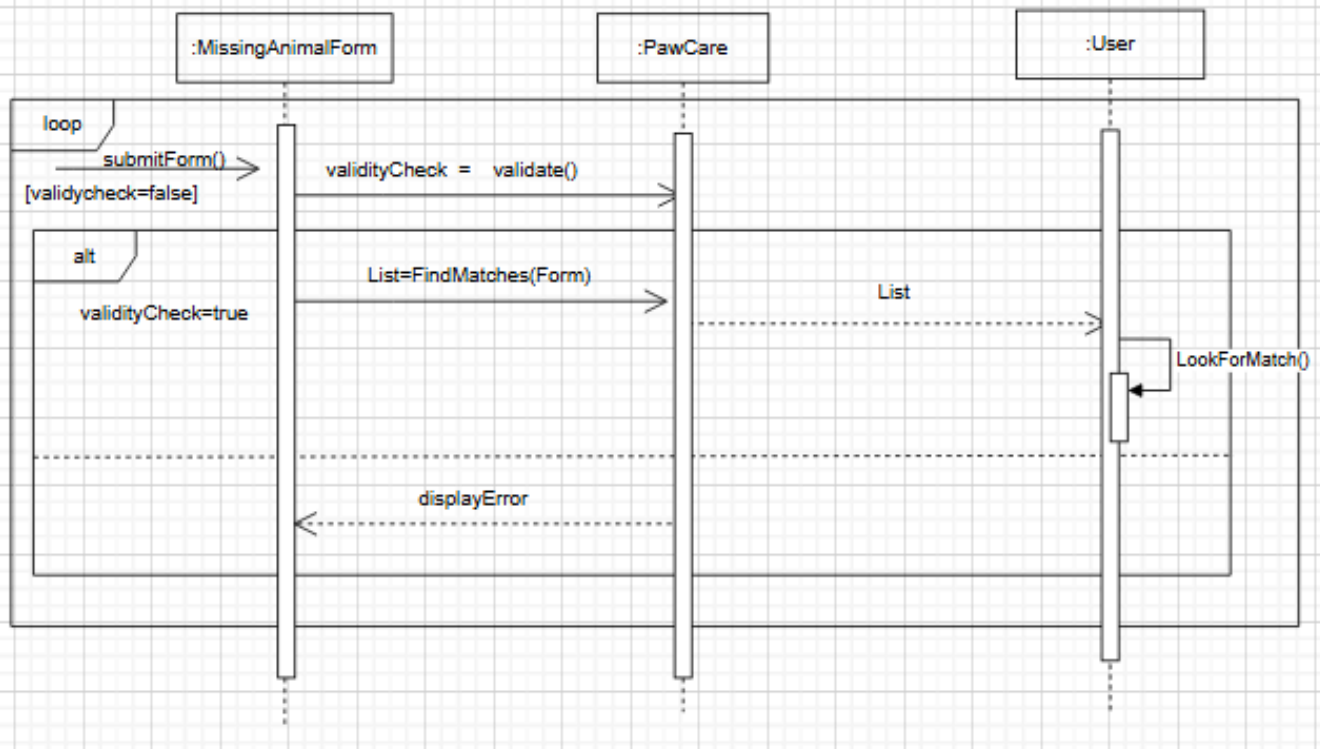
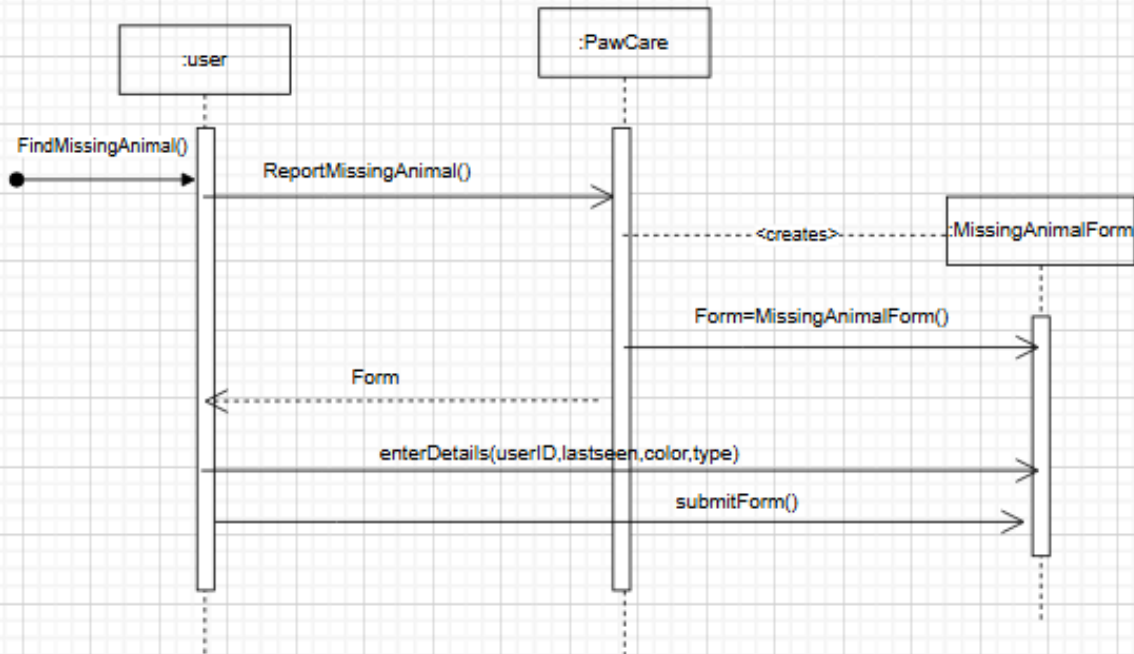


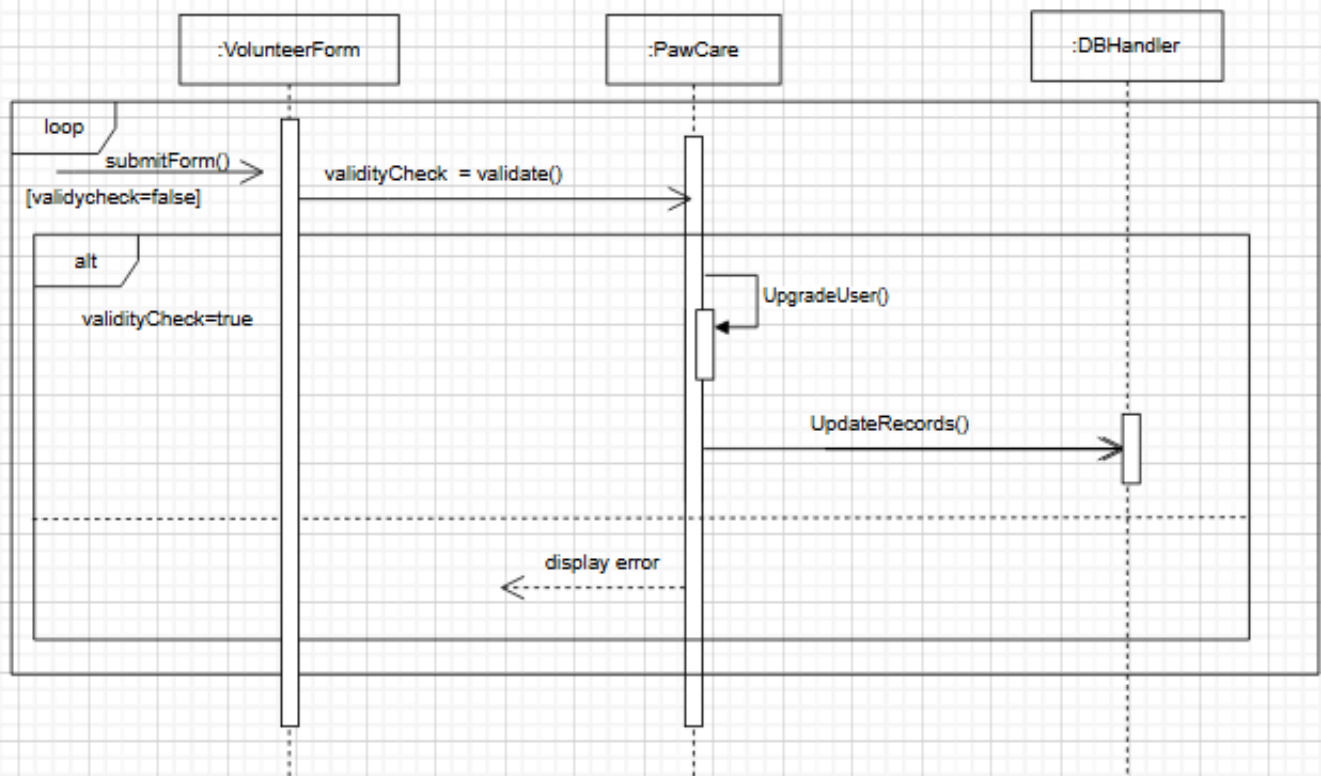
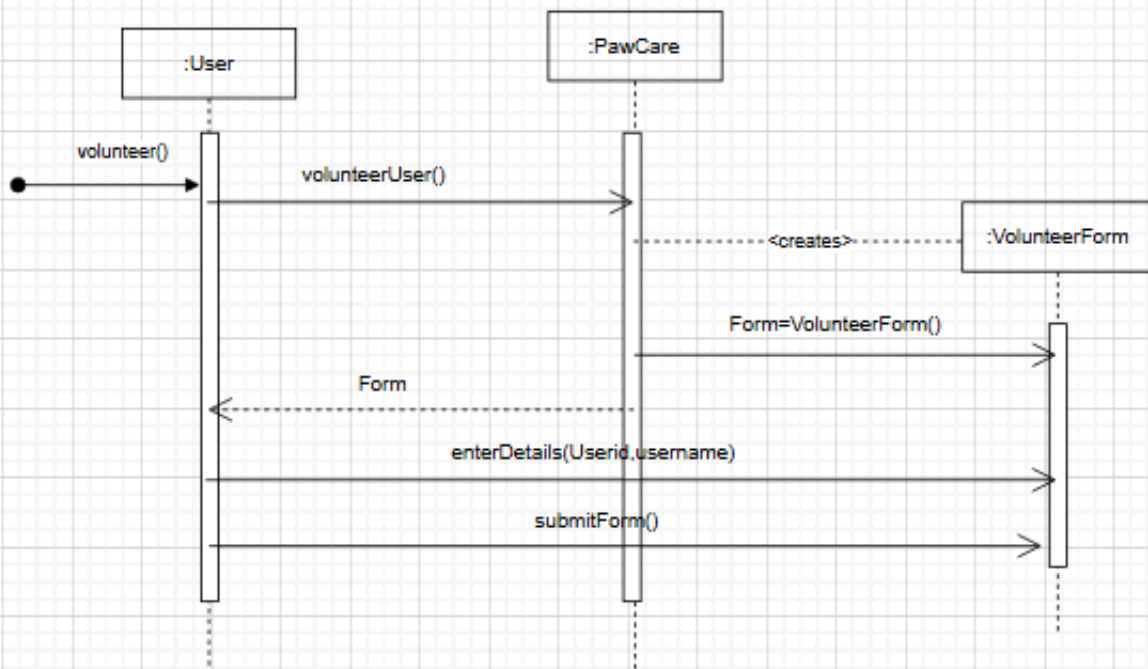
Use case: Put up for adoption

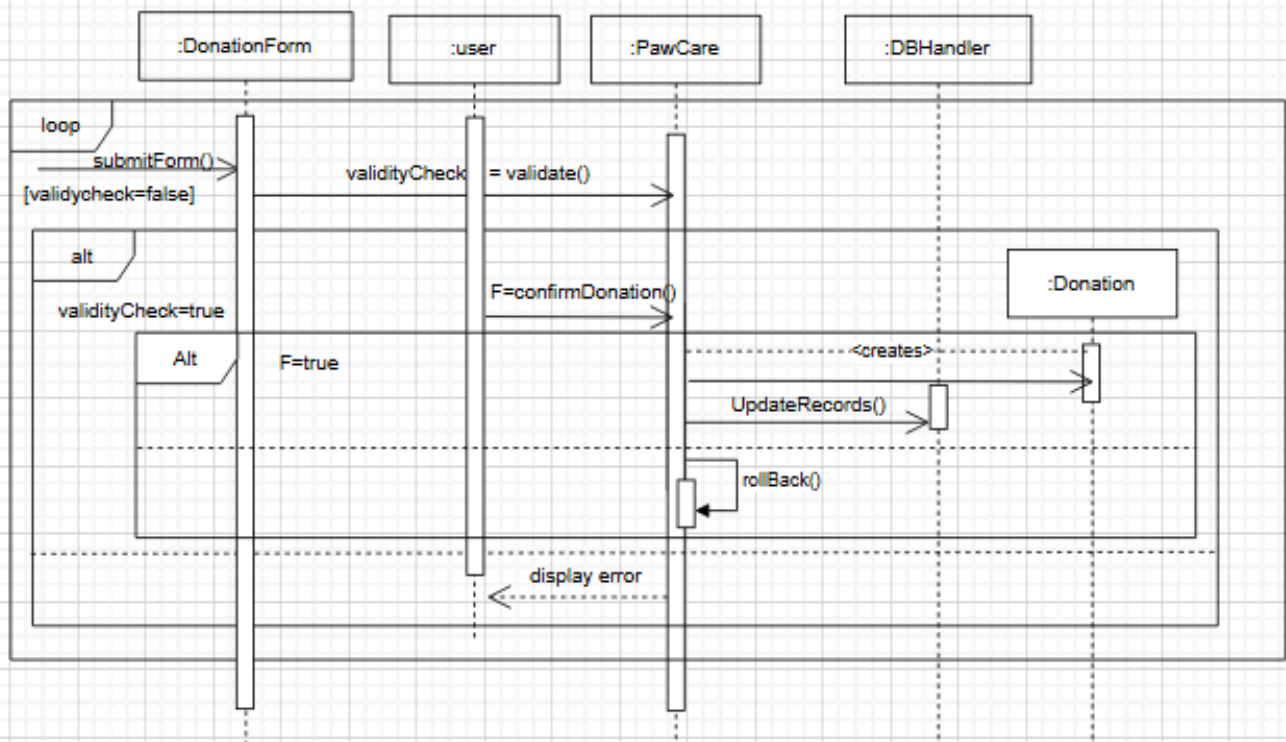
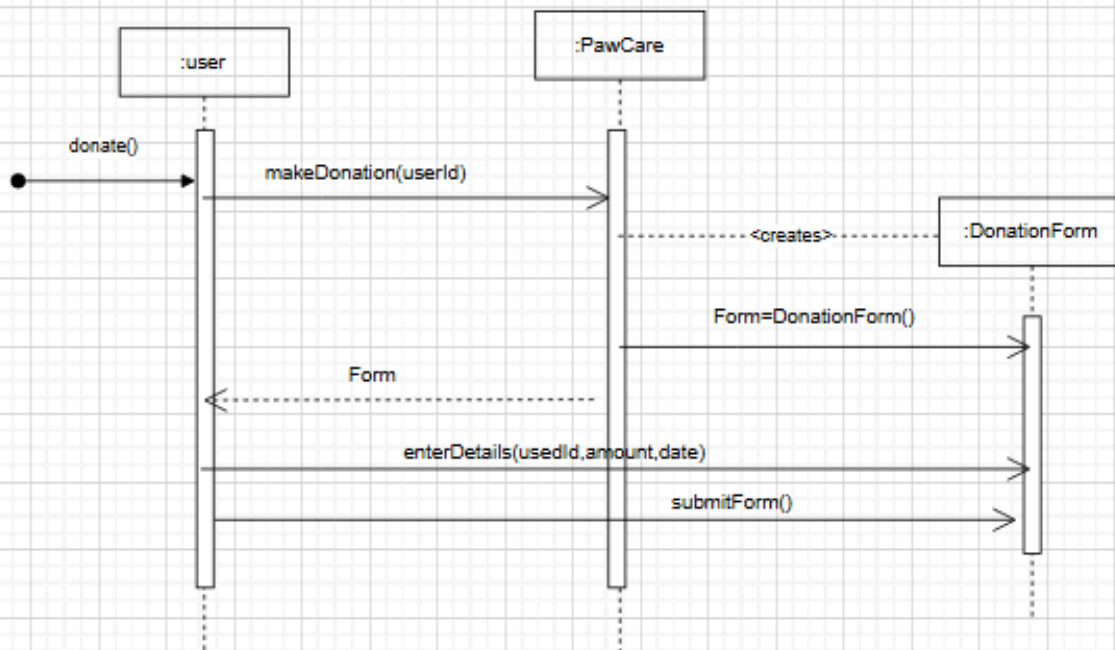


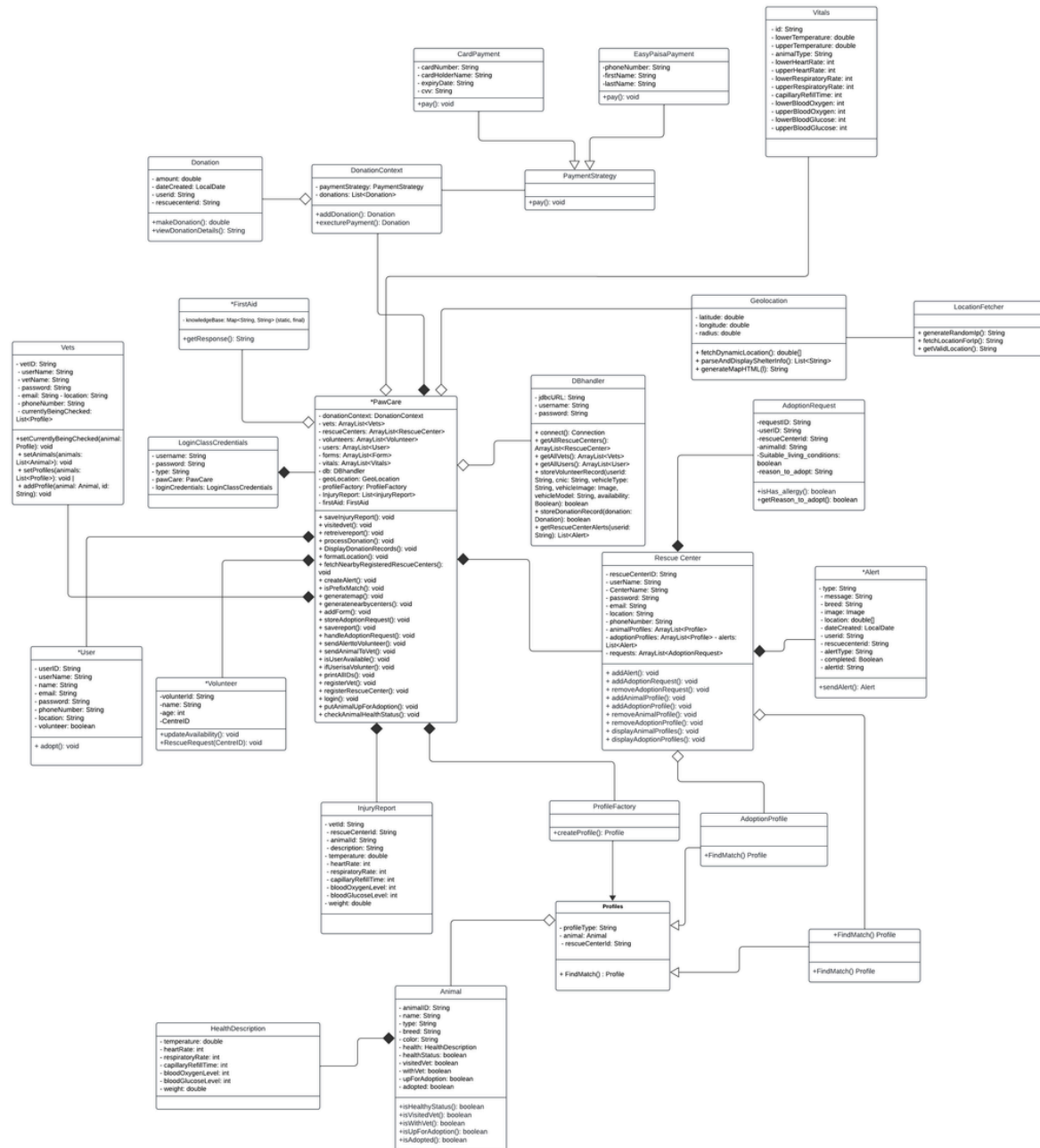
Use case: Adopt Animal

Use case: Manage animal records

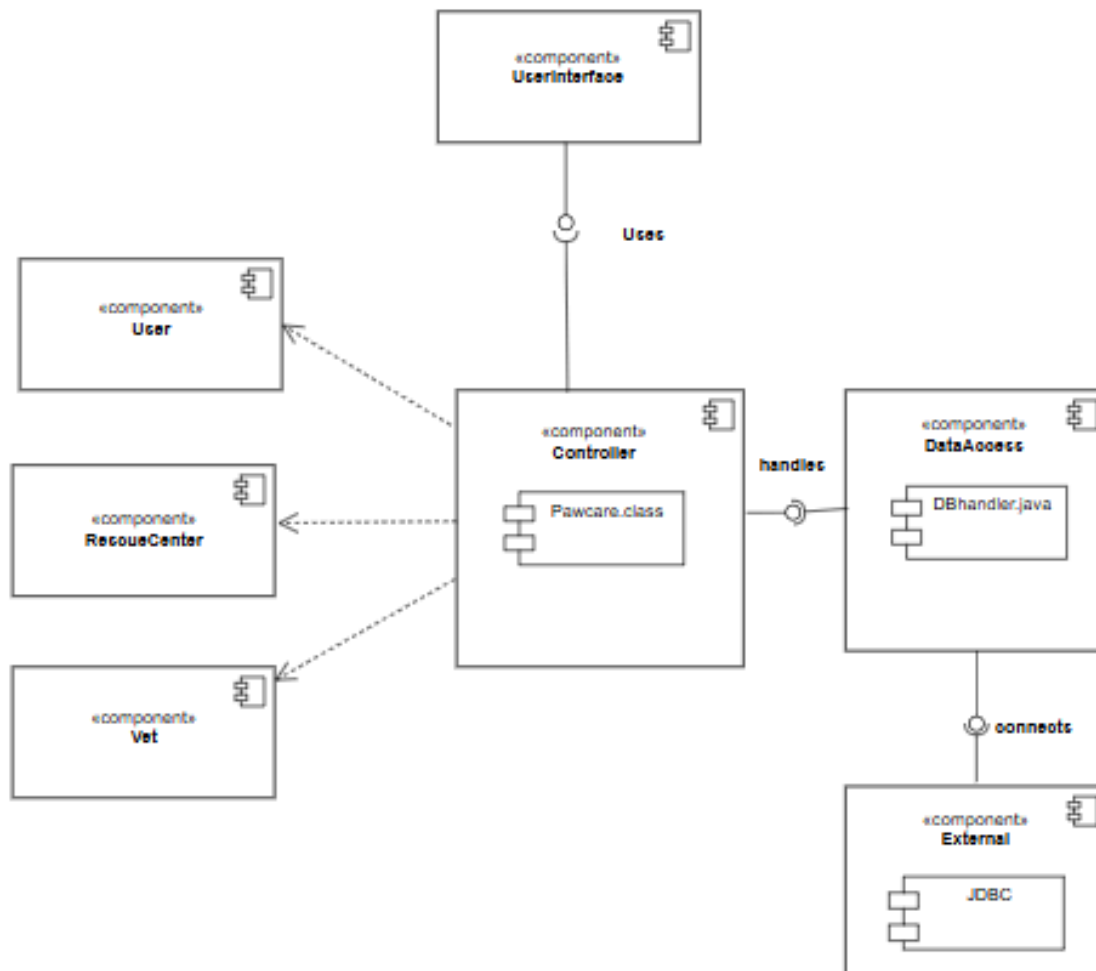
Use case: Report missing animal

Use case: Volunteer for rescue

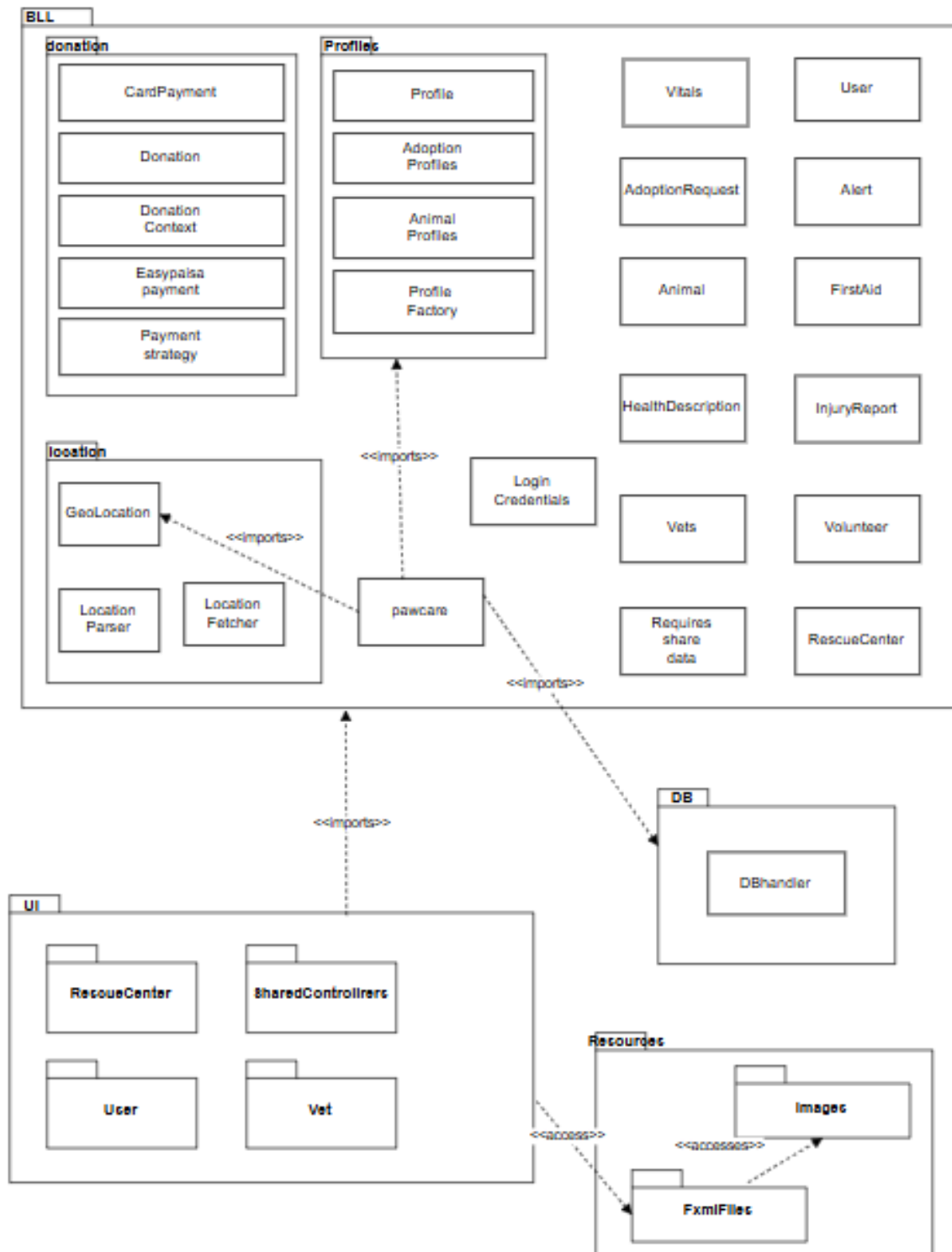
Use case: Donate



8. Component Diagram



9. Package Diagram



10. Deployment Diagram

